Asteroid Watch

Documentation set, October 9, 2026

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Part 1: Overview

`Asteroid Watch/README.md`

Asteroid Watch

Status: draft. Not live yet. Phase A (the Daily Asteroid Brief) is not built. Last updated: Oct 9, 2026.

What it is

Asteroid Watch shows daily close approaches and fireballs from NASA's Jet Propulsion Laboratory (JPL), shows how each asteroid's predicted pass changes over time, and publishes the Daily Asteroid Brief: an installable web app with a plain-language brief each morning, an archive, an RSS feed and, in Phase B, an optional email edition.

The full plan is in requirements.md and design-doc.md. The platform it runs on is documented in ../Platform/.

Live link

None yet. The link is added when Phase A is public.

What you need before running anything

These are the accounts and keys the project needs. None of them is stored in this repository.

Need	Where it is used	Phase	Where the secret lives
AWS account with the platform's cloud base (buckets, Glue, Lambda, API Gateway)	Everything	A	Platform step 0.2
JPL close-approach and fireball APIs	Daily ingest	A	No key is required by the public API pages (checked Oct 9, 2026)
Model API key and a $15 monthly spend limit set in the provider account	Batch brief writing and the clarity judge	A	Parameter Store (platform ADR-0007)
Amazon SES with verified test addresses (sandbox)	Email edition	B	Parameter Store
Keyed-hash secret for subscriber codes	Nightly subscriber copy	B	Parameter Store (ADR-0017)
Snowflake trial account	Marts and history questions	B	Parameter Store

Setup

The setup commands are added when the pipelines exist. Until then this section stays short on purpose: the design doc and the platform setup in ../Platform/ describe what is built and in what order.

Tests

Planned, not yet written. The design doc (section 13) lists them:

- Unit tests on small sample JSON for the parsers, the signature check, the pull key, the number verifier (a planted wrong number must block a brief), the keyed hash and the RSS builder.
- Spark tests on sample files, run on GitHub's free CI machines (platform ADR-0030): a run twice gives the same counts, and a query as of a past pull returns that pull's values.
- Front-end tests for the time conversion on the brief page.

Where to find things

Document	What it holds
requirements.md	Features, functional requirements, quality scenarios, acceptance criteria
design-doc.md	Architecture, components, data model, API, runtime and deployment, open questions (section 16)

Document	What it holds
adrs/	One decision per file, including the defaults that John has not yet confirmed
contracts/	Data contracts for each source and each published table
openapi.yaml	The API for the brief, the close-approach and fireball routes, and the subscription routes
operations.md	Freshness, monitoring, runbooks, cost
model-card.md	The model's use, its limits and its evals
security-and-privacy.md	Threat model, personal data and its deletion periods, licences
changelog.md	What changed in each revision

Credits

Close-approach and fireball data come from NASA/JPL. The credit wording and the terms are not confirmed yet (open question OQ-24 in the design doc). Do not publish the credit line until that is checked.

Part 2: Requirements

`Asteroid Watch/requirements.md`

Asteroid Watch. Requirements

Status: draft for John's review. Last updated: Oct 9, 2026. Owner: John.

Source labels used below: BP = Build Plans for the Five Projects (Oct 8, 2026); BPD = Build Plan Decisions (Oct 9, 2026); DP = Documentation Plan (Oct 9, 2026). Step numbers (2A.1 and so on) are the build plan's steps. ADR numbers point to files in adrs/. Platform documents do not exist yet and are linked by placeholder path (`../Platform/...`). "Done when" means the build plan's check column for a step. Technical terms (Airflow, DAG, Spark, Iceberg, partition, dbt, mart, session box, spot, Parameter Store and others) are explained in the glossary, design-doc.md section 17. "The platform" means the shared AWS base, CI and tooling built before any project. The "session box" is the temporary AWS server that runs the nightly pipelines; the public site does not depend on it.

Rule for this document: every value is a recorded decision (with its ADR, the architecture decision record, or its source) or a confirmed fact from the three source files. Anything not determined is listed in design-doc.md section 16 (Open questions) and referenced here as OQ-n.

This document follows the requirements and quality-scenario practice of arc42 sections 1 (introduction and goals) and 10 (quality requirements).

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1. Introduction and goals

1.1 What the project is

Asteroid Watch shows daily close approaches and fireballs, shows how each asteroid's predicted pass has changed over time, and publishes the Daily Asteroid Brief: a small installable web app with a plain-language brief each morning (BP, Project 2). The brief was kept and expanded into its own installable web app because John found it interesting (BPD, Projects and scope; ADR-0001).

1.2 Users and their needs

The build plan names no personas. The groups below are derived only from the features the plan lists. Each need is worded as the plan words it.

Group	Derived from	Need
Brief reader on a phone or computer	2A.3	A plain-language brief each morning, a page per day, an archive, the closest approaches, the day's fireballs on a map, and which predictions changed since yesterday. The app installs on a phone.
Reader who wants a simpler text	2B.2	An "explain it simply" reading level in addition to the standard one.
Feed-reader user	2B.2	An RSS feed of the brief.
Email subscriber	2B.2	An optional email edition with double opt-in and one-click unsubscribe.
Curious reader of prediction history	2B.1, 2B.3	A per-asteroid chart of how the predicted pass changed, flags where a predicted distance moved between pulls with an explanation, and answers to "what did we believe on a given date?".
Maintainer (John)	2A.1, 2B.4, the build plan's 16 engineering standards	Raw data that can be replayed and audited, an alert when the JPL API changes, a dashboard of usage across the five portfolio sites, and a status page entry for each pipeline.
Project 4 (another project in the portfolio, which uses Project 2's data)	4B.3	The fireball table from Project 2 as the baseline for Project 4's backtest (its comparison of scores with known shower peaks).

1.3 Quality goals (top)

1. Numbers in the brief are always in the tables (2A.3).
2. The public site answers while the session box is off (BP, Platform skeleton and 2A.3; ADR-0002).
3. A rerun never changes the data (2A.2; ADR-0003).
4. Cost stays under the plan limits: under $50 a month overall, a $15 monthly model spend limit, cheapest model only (BPD, Cost, limits and operations).
5. Real email addresses never leave DynamoDB (2B.2; ADR-0007).

1.4 Stakeholders

Stakeholder	Interest
John (owner, reviewer, sole maintainer)	Approves each document before coding of the part begins (DP, How the documents get drafted).
Readers, subscribers	See 1.2.
Data and service providers (JPL, the NASA Jet Propulsion Laboratory, for close approaches and fireballs; Amazon SES for mail)	Their terms of use and credit lines. JPL terms are not recorded in the source files: OQ-24.

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2. Features by phase

Phase	Feature	Build plan step	Summary
A	F-A1 Ingest	2A.1	Airflow pulls JPL close-approach and fireball JSON daily; `dist-max=1LD`; signature version saved with every response; alert when it changes from the version pinned for that API (ADR-0015).
A	F-A2 Versioned storage	2A.2	Spark writes each pull to its own Iceberg partition keyed by pull time and content hash.
A	F-A3 Daily Asteroid Brief	2A.3	Installable web app (PWA), page per day, archive, batch-written plain-language brief, every number checked against the tables, no model calls on brief pages.
B	F-B1 History and warehouse	2B.1	Date-range backfill, dbt SCD Type 2 prediction-history model, per-asteroid chart, marts on Snowflake.
B	F-B2 Brief upgrades	2B.2	Two reading levels, RSS feed, optional SES email edition with double opt-in and one-click unsubscribe, shareable image card, LLM judge calibrated on 30 hand-graded briefs.
B	F-B3 Prediction shifts	2B.3	Statistical rule flags asteroids whose predicted distance moves between pulls; the model explains each flag; history questions go through the metrics tools with logged SQL fallback.
B	F-B4 Portfolio analytics	2B.4	Usage events from all five sites (no raw IP addresses) and CloudFront logs load into Snowflake; dbt canonical tables; MetricFlow metrics; Superset dashboard with nightly public snapshot charts.
C	F-C1 Impact-risk list	Phase C	Add the Sentry impact-risk list once its terms are checked.
C	F-C2 Push notifications	Phase C	Push notifications when a notably close pass is coming.

Phase C starts only after every project has reached Phase B (BP, How the plan works). Phase C is in the second round and is not detailed beyond the plan's two items.

First-run checks (BP): Phase A: `dist-max=1LD` returns a known past close approach. Phase B: a test email sends from Amazon SES's sandbox mode.

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3. Functional requirements

IDs are new in this document. Each requirement restates text from the build plan or a decision; the source column shows where.

Phase A

ID	Requirement	Source
FR-1	The ingest DAG (an Airflow job that runs on a schedule) pulls the JPL close-approach JSON and the JPL fireball JSON once a day.	2A.1
FR-2	The ingest DAG requests close approaches with the parameter `dist-max=1LD`, which limits results to passes closer than the Moon.	2A.1; ADR-0005
FR-3	The ingest DAG stores every raw response with its hash, load time and the API's signature version.	2A.1; BP rule 1 (keep raw files immutable, with source URL, fetch time and hash)
FR-4	The ingest DAG raises an alert when an API's signature version differs from the version pinned for that API: 1.5 for close approaches and 1.2 for fireballs.	2A.1; ADR-0004; ADR-0015 (the 1.2 value was read from the live API on Oct 9, 2026)
FR-5	The pull-writing Spark job writes each pull to its own Iceberg partition, keyed by pull time and content hash.	2A.2; ADR-0003
FR-6	When a pull is rerun, the Spark job overwrites only that pull's partition.	2A.2; BP rule 3 (tasks are idempotent and atomic, so overwrite one partition, use the run's data interval, and never blind-append)
FR-7	The Spark job does not count identical content as a new version.	2A.2
FR-8	A query that names a past pull time ("as of" that pull) returns the values of that pull.	2A.2
FR-9	The publish gate (a set of Great Expectations checks) runs before publishing and checks freshness, empty results and value ranges.	BP technology table (Great Expectations, Project 2: A); BP rule 4 (write, audit, publish, so new data lands in staging and goes live only after the checks pass). The plan lists no checks for Project 2. Project 1 step 1A.5 names three checks (freshness, empty results and value ranges), and Project 2 adopts the same three. The thresholds are open: OQ-7.
FR-10	The web app provides one page for each day and an archive of past days.	2A.3
FR-11	The brief job (a batch job in the nightly run) writes the brief each morning in plain language from the tables. The brief covers the closest approaches, the day's fireballs on a map, and which predictions changed since yesterday.	2A.3
FR-12	The brief job checks every number in a brief against the tables before publishing, and blocks any brief that contains a number that is not in the tables.	2A.3; ADR-0002
FR-13	Opening a brief page makes no model call. (The public history questions of FR-32.3 are the only public model use, from Phase B.)	2A.3; ADR-0002 (amended, Decided by John, Oct 9, 2026)
FR-14	The app can be installed on a phone as a progressive web app (PWA).	2A.3; ADR-0001
FR-15	The brief job publishes today's brief without manual action each morning, after the nightly run.	2A.3; BPD defaults (nightly run 10:00 UTC)
FR-16	The pipelines store every timestamp in UTC; the web page converts it to the reader's local time.	BP rule 15 (store every timestamp in UTC and convert only on the page); BPD Data storage
FR-17	The web page shows the cold-start notice (the first view after a quiet period can be slow): "The first request after a period of inactivity may take up to 10 seconds while the service initializes."	BP step 0.4 (serving template)

ID	Requirement	Source
FR-18	Each DAG writes freshness and row counts to the metrics table, so that the status page shows each pipeline's last good run. The platform sends an email when a run fails or a freshness target is missed.	BP step 0.5
FR-19	The web page credits every source it uses and carries a terms note for it.	The build plan does not state this for Project 2. It follows the pattern of Projects 1 and 4 (steps 1A.6 and 4A.1), where each source gets a terms note and a credit line. JPL terms are not recorded: OQ-24.

Phase B

ID	Requirement	Source
FR-20	The backfill job (a job that loads past data) loads close approaches and fireballs for a date range with no gaps or duplicates.	2B.1
FR-21	dbt builds an SCD Type 2 prediction-history model (a table that keeps every past version of each row) and the data for a per-asteroid chart.	2B.1; ADR-0008
FR-22	The dbt marts (final tables for reports) run on Snowflake, which reads the Iceberg tables through the Glue catalog. Airflow runs a refresh step at the end of each pipeline run so that Snowflake sees the new data.	2B.1; BP Snowflake decision and "Snowflake setup"
FR-23	The brief job produces each brief at two reading levels: standard and "explain it simply".	2B.2; ADR-0011
FR-24	The brief job publishes an RSS feed (a feed that feed-reader apps can poll) of the brief.	2B.2
FR-25	The email sender sends an optional email edition through Amazon SES. A person receives it only after confirming the address (double opt-in), and every email has a one-click unsubscribe. For now the edition stays in the SES sandbox, so only verified test subscribers receive mail.	2B.2; ADR-0006 (Decided by John, Oct 9, 2026)
FR-26	The subscription handlers store sign-ups in DynamoDB.	2B.2; ADR-0007
FR-27	The nightly copy job reads the sign-ups by scan and replaces each email address with an HMAC-SHA-256 code (a keyed hash: a code that cannot be reversed without a secret key). The key is in Parameter Store, the AWS place for secrets. Real addresses never leave DynamoDB.	2B.2; BPD Data storage; ADR-0007; ADR-0017
FR-28	The brief job produces a shareable image card for each brief.	2B.2
FR-29	An LLM judge (a language model that scores text) scores brief clarity. The judge is calibrated on 30 briefs graded by hand, and the clarity scores are published.	2B.2; ADR-0011
FR-30	The shift-detection job applies a statistical rule that flags asteroids whose predicted distance moves between pulls.	2B.3; ADR-0009
FR-31	The `ShiftExplainer` (design doc C15) writes an explanation for each flag, in the nightly batch run.	2B.3; ADR-0009
FR-32.1	The history-question route answers questions such as "what did we believe on a given date?" through the metrics tools (predefined, named calculations that the MCP metrics server provides; the route, the metrics tools and the MCP metrics server are one chain).	2B.3; BPD Agent queries; 1B.5 pattern
FR-32.2	If no metric fits a question, the route runs a read-only SQL query from an allowed list instead, and logs that query separately.	2B.3; BP 1B.5
FR-32.3	The route is public, with a daily cap, a token limit per answer and abuse limits (numbers open: OQ-17).	Decided by John, Oct 9, 2026, so that readers can ask what was believed on a past date; ADR-0002, ADR-0009
FR-33	The analytics loader loads usage events and CloudFront logs into Snowflake, with no raw IP addresses. The platform defines the event format and collection point; Project 2 adds each site as that site ships.	2B.4; ADR-0010 (Decided by John, Oct 9, 2026)

ID	Requirement	Source
FR-34	dbt builds canonical tables: sessions, lookups, answers, feedback, model calls.	2B.4
FR-35	MetricFlow defines weekly users, lookup success, answer accuracy and cost per answer.	2B.4
FR-36	Each mart has a written freshness target (the maximum age its data may reach), written during the build.	2B.4
FR-37	Superset shows the dashboard while the session box is on. The nightly run publishes snapshot charts (fixed images of the charts) to the public site.	2B.4; BP tools table
FR-38	Every metric has a written definition.	2B.4
FR-39	OpenLineage (lineage events) and the dbt docs catalog (a generated page describing the tables) cover the Project 2 pipelines.	BP technology table (OpenLineage and dbt docs, Project 2: B)
FR-40	The sign-up and unsubscribe handlers write only to always-on storage (DynamoDB). The nightly run picks the data up later.	BP rule 16 (anything that runs 24/7 writes only to always-on storage)

Phase C

ID	Requirement	Source
FR-41	The project adds the Sentry impact-risk list (a list of asteroids with a possible impact risk) once its terms have been checked.	Phase C
FR-42	The app sends a push notification when a notably close pass is coming ("notably close" is not defined: OQ-23).	Phase C

All phases

ID	Requirement	Source
FR-43	When the nightly run is late or has failed, each web page shows the old data with a visible late notice and the time of the last good run, and does not hide the data. This rule applies to every project.	Decided by John, Oct 9, 2026, so that readers still see the last good data and are told it is late; ADR-0013

Constraints that apply to every functional requirement

- The build plan's 16 engineering standards (rules for pipelines, such as idempotent tasks, UTC timestamps and least privilege) apply from the first phase that touches them. Each rule is explained at its first use below.
- Dev and live are fully separate; dev cannot write live data (BPD Data storage).
- Every hard limit on cost items applies: a maximum number of rows and a maximum runtime per Spark job, with the exact values set from the measured size estimate (BP, Hard limits). The values for Project 2 are not yet set: OQ-6.

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4. Quality requirements (arc42 section 10 style)

Each quality requirement has a measure or scenario source and the way it is verified. Where the plan sets no number, the requirement states the behavior and the number is listed as open.

4.1 Quality tree

Quality	Sub-quality	IDs
Correctness	Numbers in text match tables	QR-1
Reliability	Idempotent reruns (a rerun gives the same result), old data never shown as current	QR-2, QR-3, QR-4
Availability	Site up while the session box is off	QR-5
Freshness	Daily pull, morning brief	QR-6
Cost	Monthly and per-run limits	QR-7, QR-8

Quality	Sub-quality	IDs
Privacy	Email and IP handling, timestamps	QR-9, QR-10
Security	Keys, least privilege, abuse	QR-11, QR-12
Maintainability and observability	Upstream change detection, status, lineage	QR-13, QR-14
Usability	Installability, clarity of text	QR-15, QR-16
Performance	Cold start, latency	QR-17
Compliance	Email opt-in, source terms	QR-18, QR-19

4.2 Quality scenarios

ID	Quality	Scenario (stimulus, environment, response, measure)	Source	How verified
QR-1	Numeric faithfulness (every number in the text is in the tables)	A generated brief contains a number that is not in the tables. The publish step blocks the brief, and nothing reaches the site.	2A.3	Unit test in CI (the automatic checks that run on every change) with a planted wrong number: publishing is blocked and the job exits with an error. In live use, a blocked brief creates a failed run and an alert.
QR-2	Idempotence	The same pull is rerun. The Iceberg row count is unchanged and only that pull's partition is rewritten.	2A.2; BP rule 3	CI test on sample files: run twice, compare counts. Matches the Done-when for 2A.2.
QR-3	Replayability	A query as of a past pull returns that pull's values. Raw responses carry hash, load time and API version.	2A.1, 2A.2; BP rule 1	CI test: write two pulls with different content, query as of the first.
QR-4	Old data never looks current	The nightly run fails or the nightly run is skipped (no spot capacity available). The page shows the old data with a visible late notice and the time of the last good run, and never hides it.	Decided by John, Oct 9, 2026 (ADR-0013); BP nightly-run rule that a skipped night is caught up next time	Front-end test: with a last-good-run time older than the threshold, the notice shows and the old brief stays visible. The threshold is OQ-9.
QR-5	Availability	The session box is off. The public app and its endpoints still answer.	BP Platform skeleton; 0.4; 2A.3	Done-when check run with the box off.
QR-6	Freshness	The nightly run starts at 10:00 UTC, is expected to take about 45 minutes, and stops at a hard maximum of 2 hours. The brief job publishes today's brief without manual action. The platform sends an email when a freshness target is missed.	BPD defaults; BP nightly run; 0.5; 2A.3	Status page shows last good run for each Project 2 DAG; Done-when "today's brief publishes on its own each morning". Numeric table freshness targets are OQ-7.
QR-7	Model spend	Brief generation uses only the cheapest model, in batch, cached. Monthly model spend stays under the $15 limit set in the API account before the first call.	BP Model API; BPD AI spending	Limit exists in the API account (checked before first call); cost per brief from the call log (`llm_calls`, platform shared table).
QR-8	Job limits	A Spark job that exceeds its row maximum or runtime maximum stops before publishing and opens an incident.	BP Hard limits	CI test with a sample that exceeds a lowered limit. Values: OQ-6.
QR-9	Personal data (email)	Real email addresses exist only in DynamoDB. The nightly copy holds only a keyed hash; the key is in Parameter Store.	2B.2; BPD Email addresses	Test on the copy job output; search of lake and warehouse tables for the address pattern returns nothing (Done-when check for 2B.4 states the same for IP addresses).

ID	Quality	Scenario (stimulus, environment, response, measure)	Source	How verified
QR-10	Personal data (IP and time)	No raw IP address reaches the warehouse. All timestamps stored in UTC.	2B.4; BP rule 15	Done-when check for 2B.4; schema test that no IP column exists.
QR-11	Secrets	No key in code or GitHub; all keys in Parameter Store; dev roles cannot write live data.	BPD Key storage, Dev and live	Gitleaks and Trivy scans in CI (platform step 0.1, repository and CI setup); a dev role's attempt to write live data is refused (platform step 0.2, cloud base, whose check this is).
QR-12	Abuse resistance	Nobody can use the subscribe endpoint or the public history-question route to run up SES or model charges, or to send mail to third parties, without a limit. History questions have a daily cap, a token limit per answer and abuse limits.	BP "Nothing in the plan can run up a surprise bill" principle; ADR-0002	Tests that an over-cap request is refused (429) once the numbers are set. Subscribe limits: OQ-14; history numbers: OQ-17.
QR-13	Upstream change detection	An API's signature version differs from its pinned version (1.5 close approaches, 1.2 fireballs; ADR-0015). The ingest DAG raises an alert.	2A.1	Unit test with a sample response whose version is not the pinned value for that API.
QR-14	Observability	Each DAG writes freshness and row counts; the status page shows the last good run; lineage covers raw to gold.	0.5; 1B.3 pattern; BP rule 11 (pipelines report freshness and row counts, with a status page and one ticket per incident)	Status page and lineage graph inspection.
QR-15	Installability	The app installs on a phone.	2A.3	Install on a phone (Done-when).
QR-16	Clarity	The LLM judge scores clarity. It is calibrated against 30 hand-graded briefs, and agreement and scores are published.	2B.2	Published score table. Pass threshold is not set: OQ-18.
QR-17	Cold start	The page shows the cold-start notice. The project measures median warm latency and the 95th-percentile cold-start time, and publishes both in the README.	0.4; 1B.9	Notice text check; measurements per platform step 1B.9. Numeric latency targets are not decided: OQ-20.
QR-18	Mail compliance	Subscription needs a confirming action (double opt-in). Every email has a one-click unsubscribe.	2B.2; ADR-0006	Done-when: test subscriber receives the email after confirming; unsubscribe test.
QR-19	Source terms	Every source has a terms note and a credit line, and the licenses of open data are respected.	BP assumptions and 4A.1 pattern	Review of page and README. JPL terms: OQ-24.
QR-20	History accuracy	History questions return correct answers.	2B.3	John checks by hand the answers to 10 history questions.
QR-21	Data integrity of backfill	A multi-year range loads without gaps or duplicates.	2B.1	Done-when check; row counts per day compared with expectation.

5. Acceptance criteria

Each feature's acceptance criteria are the build plan's "Done when" checks, quoted or lightly shortened. A feature is accepted when every listed check passes.

F-A1 Ingest (2A.1)

- AC-A1.1: Raw responses land with hash, load time and API version. (Done when, 2A.1)
- First-run check: `dist-max=1LD` returns a known past close approach. The test case is 2020 QG (August 16, 2020), named by John's decision and found in the live API on Oct 9, 2026 (ADR-0005, ADR-0014). The fallback if it fails is OQ-2.
- Linked requirements: FR-1 to FR-4, QR-3, QR-13.

F-A2 Versioned storage (2A.2)

- AC-A2.1: Rerunning a pull leaves the row count unchanged, and a query as of a past pull returns its values.
- Linked requirements: FR-5 to FR-8, QR-2, QR-3.

F-A3 Daily Asteroid Brief (2A.3)

- AC-A3.1: The app installs on a phone, today's brief publishes on its own each morning, and a brief with a number not in the tables is blocked.
- Linked requirements: FR-9 to FR-19, QR-1, QR-4 to QR-6, QR-15.

F-B1 History and warehouse (2B.1)

- AC-B1.1: A multi-year range loads without gaps or duplicates, and one asteroid's changing prediction is charted.
- Linked requirements: FR-20 to FR-22, FR-39, QR-21.

F-B2 Brief upgrades (2B.2)

- AC-B2.1: The RSS feed validates, a test subscriber receives the email after confirming, and clarity scores are published.
- First-run check: a test email sends from Amazon SES's sandbox mode.
- Linked requirements: FR-23 to FR-29, FR-40, QR-9, QR-12, QR-16, QR-18.

F-B3 Prediction shifts (2B.3)

- AC-B3.1: Flags appear with explanations, and 10 history questions return answers that John has checked by hand (the build plan words this as "answers you checked by hand").
- Linked requirements: FR-30 to FR-32.3, QR-20.

F-B4 Portfolio analytics (2B.4)

- AC-B4.1: The weekly dashboard loads, the public snapshot updates nightly, every metric has a written definition, and no raw IP address reaches the warehouse.
- Linked requirements: FR-33 to FR-38, QR-10.

F-C1 and F-C2 (Phase C)

- The build plan gives no Done-when checks for Phase C items. Phase C gate (BP, Second round): the design note is updated with measured numbers. Acceptance criteria for F-C1 and F-C2 are therefore open until Phase C planning (OQ-23).

Phase gates (BP, Build order and checkpoints)

- Project 2 Phases A and B gate: the Daily Asteroid Brief publishes on its own each morning; then history, the email edition and portfolio analytics.

---

6. Traceability

Feature	Requirements	Quality	ADRs	Interface specs
F-A1	FR-1 to FR-4	QR-3, QR-13	0004, 0005, 0014, 0015, 0018	contracts/src_jpl_close_approach.yaml, contracts/src_jpl_fireball.yaml

Feature	Requirements	Quality	ADRs	Interface specs
F-A2	FR-5 to FR-8	QR-2, QR-3	0003, 0016	contracts/close_approach_pull.yaml, contracts/fireball_pull.yaml
F-A3	FR-9 to FR-19, FR-43	QR-1, QR-4 to QR-6, QR-15, QR-17	0001, 0002, 0012, 0013	contracts/gold_brief_tables.yaml, openapi.yaml
F-B1	FR-20 to FR-22, FR-39	QR-21	0008	contracts/prediction_history.yaml
F-B2	FR-23 to FR-29, FR-40	QR-9, QR-12, QR-16, QR-18	0006, 0007, 0011, 0017, 0019	contracts/src_subscriber_signups.yaml, contracts/subscriber_lake_copy.yaml, openapi.yaml
F-B3	FR-30 to FR-32.3	QR-12, QR-20	0002, 0009	contracts/prediction_flag.yaml
F-B4	FR-33 to FR-38	QR-10	0010	contracts/src_usage_events.yaml, contracts/src_cloudfront_logs.yaml, contracts/portfolio_analytics_marts.yaml

Part 3: Design document

`Asteroid Watch/design-doc.md`

Asteroid Watch. Design doc

Template: the 17-section design doc template of the Documentation Plan (Google design doc, arc42, C4 model).

Source labels: BP = Build Plans for the Five Projects (Oct 8, 2026); BPD = Build Plan Decisions (Oct 9, 2026); DP = Documentation Plan (Oct 9, 2026). "Step 2A.1" and similar are BP steps. Platform documents do not exist yet and are linked by placeholder path (`../Platform/...`) and not repeated here.

How names in this document were set. The sources name steps, tools and behaviors but not table, DAG, class or endpoint names. Every name for a table, DAG, module, class, function or URL path in this document is a proposal made here so that engineers have something to build from; it is not a recorded decision. John's approval of this document approves the names (OQ-22). Values (numbers, versions, limits) are never proposed: they are either cited or listed in section 16.

---

1. Status, owner, reviewers, last updated

Field	Value
Project	Project 2: Asteroid Watch (including the Daily Asteroid Brief)
Status	Draft, awaiting John's review. Coding of Project 2 does not start until John approves (DP, How the documents get drafted).
Owner	John
Reviewers	John (the only reviewer named in the sources)
Last updated	Oct 9, 2026
Companion documents	requirements.md; adrs/; contracts/; openapi.yaml; platform documents `../Platform/design-doc.md`, `../Platform/adrs/` (placeholders); Project 1 `../Will I have signal out there/design-doc.md` (placeholder) for the shared map, metrics tools and eval harness
Build position	Project 2 is built fourth: Platform skeleton, Project 1, Project 3, Project 5, then Project 2, then Project 4 (BP, How the plan works).

---

2. Summary

Asteroid Watch pulls the JPL close-approach and fireball JSON every day. It keeps every pull as its own versioned Iceberg partition (a stored slice of a table), so that the predicted pass of each asteroid can be compared over time. It publishes the Daily Asteroid Brief, an installable web app (PWA) with a page per day, an archive and a plain-language brief. A batch job writes each brief every morning and checks every number against the tables before publishing, and brief pages make no model calls. Phase B adds a multi-year backfill (loading past data), a dbt SCD Type 2 prediction-history model with marts on Snowflake, two reading levels, an RSS feed, an optional Amazon SES email edition with double opt-in, a shareable image card, prediction-shift flags with explanations, public history questions through the metrics tools with a daily cap, a token limit and abuse limits, and portfolio-wide usage analytics built on the platform's event format and collection point. The outcome is a public site that stays up while the session box (the temporary AWS server that runs the nightly pipelines) is off, and a brief that publishes on its own each morning.

---

3. Context, scope and constraints

3.1 Background facts (from the sources)

- The JPL close-approach API is called with `dist-max=1LD`, which limits results to passes closer than the Moon. The ingest saves the API's signature version (the version the API reports in each response) with every response and raises an alert when it differs from the version pinned for that API: 1.5 for close approaches and 1.2 for fireballs (BP 2A.1; ADR-0004, ADR-0005, ADR-0015).
- First-run check, Phase A: `dist-max=1LD` returns a known past close approach. The test case is 2020 QG (August 16, 2020), named by John's decision (answer 39) and found in the live API on Oct 9, 2026 (ADR-0005, ADR-0014). First-run check, Phase B: a test email sends from Amazon SES's sandbox mode (BP).
- Decided by John, Oct 9, 2026: when a nightly run is late or has failed, a page shows the old data with a visible late notice and the time of the last good run, and does not hide the data (ADR-0013).
- Decided by John, Oct 9, 2026: the email edition stays in the SES sandbox for now, with verified test subscribers only (ADR-0006).
- Decided by John, Oct 9, 2026: history questions are public from Phase B, with a daily cap, a token limit and abuse limits (ADR-0002, ADR-0009).
- Decided by John, Oct 9, 2026: the platform defines the usage-event format and collection point; Project 2 builds the Snowflake tables and reports and adds each site as it ships (ADR-0010).
- Decided by John, Oct 9, 2026: a pulls log (one row per pull) plus a distinct-versions table (one data partition per distinct content) lets the partition key be pull time and content hash while identical content still does not count as a new version (ADR-0003).
- The brief is an installable web app with an RSS feed and an optional email edition (BPD; ADR-0001).
- The nightly run starts at 10:00 UTC (3 a.m. Arizona, 6 a.m. US Eastern), usually takes about 45 minutes, and stops at a hard maximum of 2 hours. It runs only on spot capacity (spare AWS servers that can be taken back at short notice). If no spot capacity is available, that night is skipped and the next run catches up (BPD defaults; BP How the box is run).
- Model API: only the cheapest model for now, $15 monthly spend limit set in the API account before the first call, content generated in batch and cached (BP Model API; BPD AI spending).
- Checked against the live JPL APIs on Oct 9, 2026: the close-approach API reports signature version 1.5 and the fireball API reports 1.2, both at `signature.version`. Each API has its own pinned version (ADR-0015).
- Defaults recorded under the security-first rule, confirmed by John on Oct 9, 2026: SHA-256 content hash of the raw bytes (ADR-0016); HMAC-SHA-256 keyed hash, with the subscriber copy read by scan and never exported (ADR-0017); a failed or malformed upstream response fails with an alert, and a signature change keeps the raw file and stops publishing (ADR-0018); 256-bit single-use confirmation tokens stored as hashes (ADR-0019).

3.2 System boundary

Inside: the Project 2 pipelines (ingest, versioned storage, gold tables, brief generation, Phase B models, subscriber copy, analytics models), the Daily Asteroid Brief web app and its API routes, the DynamoDB sign-up table and the SES sending path.

Outside (platform; "the platform" means the shared AWS base, CI and tooling built before any project): AWS accounts, S3 buckets, Glue catalog, Terraform, CI, the session box with k3s, Airflow and Spark installation, Parameter Store, status page and metrics table, alerting, the serving template (React site, map component, FastAPI, a Python web framework, on Lambda behind API Gateway), the eval harness, MCP metrics server, `llm_calls` logging, Snowflake account and resource monitor, Marquez. Links: `../Platform/design-doc.md` (placeholder).

Outside (other projects): Project 1 supplies the map component, the MCP metrics server pattern and the eval harness. Project 4 consumes the fireball table (BP 4B.3).

External systems: JPL close-approach API and fireball API; Amazon SES; the Model API (the provider is not named in the sources); Snowflake; CloudFront and S3.

3.3 Budget

The whole plan stays under $50 a month, and John works from a browser only (BPD). Project 2's own cost lines: the nightly run share (about 22 hours a month for the whole plan; BP), the model API ($15 limit shared across the plan; cheapest model), Snowflake (3 credits a month cap shared), DynamoDB (free tier), Lambda (free tier: 1 million requests and 400,000 GB-seconds a month), CloudFront (free up to 1 TB and 10 million requests a month). SES cost is not in the plan (OQ-13). No per-project budget line exists in the sources for the shared limits (model spend, Snowflake credits, storage), so Project 2 has no allocated share (OQ-6).

3.4 Licenses and terms

- JPL data terms and credit wording are not in the sources (OQ-24). On Oct 9, 2026 the close-approach API page was read: it states no terms, license or credit line. JPL's and NASA's policy pages returned HTTP 404 and were not read, so the terms are unverified and no wording is given here.
- The plan's list of non-commercial or share-alike data (Global Fishing Watch, Open-Meteo, Ookla, Databricks Free Edition, OpenCelliD) does not include JPL data. No JPL license statement is recorded either way.
- Sentry impact-risk list (Phase C): terms to be checked first (BP Phase C).

3.5 Privacy rules

- Email addresses stay in DynamoDB; elsewhere only a keyed-hash code (ADR-0007; BPD).
- No raw IP address reaches the warehouse (ADR-0010; BP 2B.4).
- All timestamps stored in UTC and converted only on the page (BP rule 15, store every timestamp in UTC and convert only on the page).
- Deletion periods for personal data are documented in the Security and privacy document written while building (DP); the Project 2 values are not decided (OQ-15).

3.6 Constraints carried from the platform

Terraform, GitHub Actions, k3s; dev and live fully separate, dev cannot write live data; every key in Parameter Store; arm64 images; Lambda and Fargate write only to always-on storage (S3, DynamoDB, CloudWatch, X-Ray); every Spark job has a maximum number of rows and a maximum runtime; the 16 engineering standards (BP).

---

4. Goals and non-goals

Goals

1. Capture every JPL pull faithfully, with hash, load time and API version, and keep each pull as its own version (2A.1, 2A.2).
2. Publish a Daily Asteroid Brief each morning with no manual action and no wrong numbers (2A.3).
3. Show how each asteroid's predicted pass changed over time and flag shifts (2B.1, 2B.3).
4. Offer the brief as an installable app, RSS and, optionally, email, at two reading levels (2B.2).
5. Provide portfolio usage analytics for all five sites without raw IP addresses (2B.4).
6. Stay live with the session box off; stay within cost limits.

Non-goals

- No model calls when a reader opens a brief page (2A.3; ADR-0002). The only public model use is history questions from Phase B, under caps (ADR-0002, ADR-0009).
- No weekly cadence: the pull is daily (BPD, Data freshness).
- No staged failures; real incidents only (BPD, Failure testing).
- No other warehouse, query engine or BI tool than the chosen ones (BP, One tool per job).
- Sentry impact-risk list and push notifications are Phase C and not designed in detail here.
- No server-side storage of exact subscriber addresses outside DynamoDB.
- No model card is written for Project 2 before coding (DP: while-building documents).

---

5. Requirements

Summary. The full list, with the source of each item, quality scenarios and acceptance criteria, is in requirements.md.

Functional (43 items, FR-1 to FR-43):

- Phase A: daily ingest with `dist-max=1LD`, raw hash/load time/API version, signature alert at 1.5; one Iceberg partition per pull keyed by pull time and content hash, idempotent reruns, as-of queries; Great Expectations publish gate; installable web app with page per day and archive; batch-written brief; number verification; no model calls on brief pages; late data shown with a late notice.
- Phase B: backfill; dbt SCD Type 2 history; Snowflake marts; two reading levels; RSS; SES email edition with double opt-in and one-click unsubscribe; DynamoDB sign-ups with keyed-hash copy; share card; LLM clarity judge calibrated on 30 briefs; statistical shift flags with explanations; history questions through the metrics tools; portfolio analytics with MetricFlow and Superset snapshots.
- Phase C: Sentry impact-risk list; push notifications.

Quality (21 scenarios, QR-1 to QR-21): numeric faithfulness (every number in the text is in the tables), idempotence, replayability, availability while the session box is off, freshness, model spend, job limits, privacy of email and IP, secrets, abuse resistance, upstream change detection, observability, installability, clarity, cold start, mail compliance, source terms, history accuracy, backfill integrity.

Acceptance: the build plan's "Done when" check for each of steps 2A.1 to 2A.3 and 2B.1 to 2B.4, listed in requirements.md section 5.

---

6. Architecture

6.1 System context (C4 level 1)

```
flowchart TB
    reader([Reader&lt;br/&gt;browser or installed app]):::person
    subscriber([Subscriber&lt;br/&gt;email edition, Phase B]):::person
    feed([Feed reader user&lt;br/&gt;RSS, Phase B]):::person
    john([John, maintainer&lt;br/&gt;alerts, hand-grading]):::person
    aw[Asteroid Watch&lt;br/&gt;Daily Asteroid Brief]:::sys
    jpl[JPL APIs&lt;br/&gt;close approach, fireball]:::ext
    ses[Amazon SES&lt;br/&gt;sends mail, Phase B]:::ext
    llm[Model API&lt;br/&gt;cheapest model, batch]:::ext
    sf[Snowflake&lt;br/&gt;marts, Phase B]:::ext
    platform[Platform&lt;br/&gt;Terraform, CI, Airflow, Spark]:::ext
    reader -->|reads briefs| aw
    subscriber -->|subscribes, confirms| aw
    feed -->|polls RSS| aw
    john -->|operates| aw
    aw -->|pulls daily| jpl
    aw -->|sends mail| ses
    aw -->|batch generation| llm
    aw -->|refresh and marts| sf
    aw -->|runs on| platform
    classDef person fill:#dbeafe,stroke:#1d4ed8,color:#0f172a
    classDef ext fill:#e5e7eb,stroke:#6b7280,color:#111827
    classDef sys fill:#1d4ed8,stroke:#1e3a8a,color:#ffffff
    classDef cont fill:#dcfce7,stroke:#15803d,color:#0f172a
    classDef store fill:#fef3c7,stroke:#b45309,color:#0f172a
```

6.2 Containers (C4 level 2)

Two diagrams show the same containers: the nightly pipeline on the session box, then the always-on serving side.

```
flowchart LR
    subgraph extsys[External systems]
        jpl[JPL APIs]:::ext
        llm[Model API&lt;br/&gt;batch calls]:::ext
        sf[Snowflake&lt;br/&gt;Phase B]:::ext
    end
    subgraph box[Session box k3s, on during the nightly run]
        airflow[Airflow DAGs&lt;br/&gt;ingest, gate, brief]:::cont
        spark[Spark jobs&lt;br/&gt;pulls, gold, copy]:::cont
```

```
        dbt[dbt project&lt;br/&gt;SCD2 and marts, Phase B]:::cont
        superset[Superset&lt;br/&gt;snapshots, Phase B]:::cont
    end
    subgraph aws[Always-on AWS]
        raw[(Raw bucket&lt;br/&gt;S3, versioned)]:::store
        lake[(Lake tables&lt;br/&gt;Iceberg, Glue)]:::store
        gold[(Published gold&lt;br/&gt;Parquet on S3)]:::store
        site[(Site bucket&lt;br/&gt;S3 and CloudFront)]:::store
        ssm[(Parameter Store&lt;br/&gt;SecureString)]:::store
    end
    airflow -->|pulls JSON| jpl
    airflow -->|writes raw| raw
    airflow -->|submits| spark
    spark -->|writes partitions| lake
    spark -->|publishes| gold
    airflow -->|batch brief calls| llm
    airflow -->|publishes verified briefs| site
    airflow -->|reads keys| ssm
    dbt -->|marts| sf
    sf -->|reads Iceberg via Glue| lake
    superset -.->|snapshot charts| site
    classDef person fill:#dbeafe,stroke:#1d4ed8,color:#0f172a
    classDef ext fill:#e5e7eb,stroke:#6b7280,color:#111827
    classDef sys fill:#1d4ed8,stroke:#1e3a8a,color:#ffffff
    classDef cont fill:#dcfce7,stroke:#15803d,color:#0f172a
    classDef store fill:#fef3c7,stroke:#b45309,color:#0f172a
```

```
flowchart TB
    reader([Reader]):::person
    subscriber([Subscriber, Phase B]):::person
    ses[Amazon SES, sandbox only]:::ext
    llm[Model API]:::ext
    subgraph box[Session box k3s, nightly run]
        airflow[Airflow DAGs]:::cont
        spark[Spark jobs]:::cont
    end
    subgraph aws[Always-on AWS]
        site[(Site bucket&lt;br/&gt;S3 and CloudFront)]:::store
        api[API&lt;br/&gt;FastAPI on Lambda with DuckDB&lt;br/&gt;behind API Gateway]:::cont
        gold[(Published gold&lt;br/&gt;Parquet on S3)]:::store
        ddb[(Sign-up table&lt;br/&gt;DynamoDB, point-in-time recovery on)]:::store
        ssm[(Parameter Store&lt;br/&gt;keyed-hash key)]:::store
    end
    reader -->|views PWA| site
    reader -->|calls data routes| api
    subscriber -->|subscribe, confirm, unsubscribe| api
    api -->|DuckDB reads| gold
    api -->|reads and writes| ddb
    api -->|history questions, Phase B, capped| llm
    api -->|confirmation mail| ses
    airflow -->|daily edition, location open OQ-13| ses
    spark -->|nightly copy, keyed hash| ddb
    airflow -->|reads keys| ssm
    classDef person fill:#dbeafe,stroke:#1d4ed8,color:#0f172a
    classDef ext fill:#e5e7eb,stroke:#6b7280,color:#111827
    classDef sys fill:#1d4ed8,stroke:#1e3a8a,color:#ffffff
    classDef cont fill:#dcfce7,stroke:#15803d,color:#0f172a
    classDef store fill:#fef3c7,stroke:#b45309,color:#0f172a
```

6.3 Components of the brief pipeline (C4 level 3)

```
flowchart LR
    gold[(Published gold&lt;br/&gt;Parquet)]:::store
    llm[Model API]:::ext
    site[(Site bucket&lt;br/&gt;S3)]:::store
    subgraph brief[Brief generation Airflow DAG]
        facts[BriefFacts builder&lt;br/&gt;Python and DuckDB]:::cont
        writer[BriefWriter&lt;br/&gt;Python, batch call]:::cont
        verifier[NumberVerifier&lt;br/&gt;Python]:::cont
        publisher[BriefPublisher&lt;br/&gt;Python]:::cont
        logger[Call logger&lt;br/&gt;platform]:::cont
    end
    facts -->|reads| gold
    writer -->|gets facts| facts
```

```
    writer -->|batch call| llm
    writer -->|logs| logger
    verifier -->|checks text| writer
    publisher -->|only if passed| verifier
    publisher -->|writes files, index, feed, card| site
    classDef person fill:#dbeafe,stroke:#1d4ed8,color:#0f172a
    classDef ext fill:#e5e7eb,stroke:#6b7280,color:#111827
    classDef sys fill:#1d4ed8,stroke:#1e3a8a,color:#ffffff
    classDef cont fill:#dcfce7,stroke:#15803d,color:#0f172a
    classDef store fill:#fef3c7,stroke:#b45309,color:#0f172a
```

Platform architecture (Terraform, CI, session box, Airflow, Spark, serving template, status page) is described in `../Platform/design-doc.md` (placeholder) and not repeated.

---

7. Component breakdown

Python package name `asteroid_watch` (proposed). Platform pieces used as they are: Airflow, Spark, `llm_calls` call logging, status metrics writer, eval harness, MCP server framework, serving template. None of the four function-level tricky parts (terrain horizon 1A.2, satellite windows 1A.3, vessel identity matching 3B.3 and 3B.4, transcript time alignment 5A.4) belong to this project; components below are therefore listed with their main functions and classes, not exhaustive function tables.

Where a function's input depends on something not yet known (JPL field names, date windows, thresholds), the table describes the input by its role, and the unknown is listed in section 16.

Phase A components

C1. JPL ingest (Airflow DAG `asteroid_watch_ingest`, module `asteroid_watch.ingest`)

Job: pull the close-approach and fireball JSON each day, record signature version, store immutable raw files.

Inputs: JPL close-approach API (with `dist-max=1LD`), JPL fireball API; the run's data interval; expected signature version for each API: 1.5 for close approaches, 1.2 for fireballs (ADR-0004, ADR-0015). Outputs: raw files in the raw bucket (each with source URL, hash, load time, API version; BP rule 1 (keep raw files immutable, with source URL, fetch time and hash)); a signature-check result; run metrics (row counts, freshness) for the status page.

Main classes and functions:

Name	Purpose	Takes	Returns
`JplClient`	Calls the two JPL endpoints	base URL settings, HTTP session	client object
`JplClient.fetch_close_approaches(date_min, date_max)`	Requests passes with `dist-max=1LD`	date window (window rule: OQ-3)	`RawResponse`
`JplClient.fetch_fireballs(date_min, date_max)`	Requests fireball records	date window	`RawResponse`
`RawResponse` (dataclass)	Holds one response	n/a	fields: source URL, HTTP status, body bytes, fetched-at (UTC), API signature version
`extract_signature_version(body)`	Reads the signature version from a response	response body	version string
`check_signature(version, expected)`	Compares with the version pinned for that API (ADR-0015) and raises an alert event on difference	version, expected	`SignatureCheck` (ok or changed)
`content_hash(body)`	SHA-256 of the raw response bytes, as 64 hex characters (ADR-0016)	body bytes	hex string

Name	Purpose	Takes	Returns
`store_raw(response, env)`	Writes the raw file and its metadata (source URL, hash, load time, API version) to the raw bucket of the right environment	`RawResponse`, environment	object key
DAG tasks `fetch_close_approaches`, `fetch_fireballs`, `check_signature`, `store_raw`, `emit_run_metrics`	Orchestration	data interval	task results

Failure handling (ADR-0018, confirmed by John on Oct 9, 2026): a non-2xx, malformed or unexpected response fails the task after retries and raises an alert. A signature change keeps the raw file and stops before any write or publish until John reviews (OQ-5). Reruns are safe because tasks are idempotent (BP rule 3, tasks are idempotent and atomic, so overwrite one partition, use the run's data interval, and never blind-append).

C2. Versioned storage (Spark job `asteroid_watch.pulls.write_pull`)

Job: parse a raw file and write it to its own Iceberg partition; never count identical content as a new version (ADR-0003).

Inputs: raw file key; content hash; data interval. Outputs: rows in Iceberg tables `close_approach_pull` and `fireball_pull` (one data partition per distinct content hash) and in the pulls logs `close_approach_pull_log` and `fireball_pull_log` (one row per pull; ADR-0003, Accepted) (contracts/close_approach_pull.yaml, contracts/fireball_pull.yaml).

Name	Purpose	Takes	Returns
`PullKey` (dataclass)	Partition key: pull time and content hash	pull time, hash	key object
`build_pull_key(data_interval, content_hash)`	Builds the key from the run's data interval, not now() (BP rule 3)	interval, hash	`PullKey`
`parse_close_approaches(raw)`	Turns JSON into typed rows (field list: OQ-1)	raw body	Spark DataFrame
`parse_fireballs(raw)`	Same for fireballs	raw body	Spark DataFrame
`write_partition(df, table, key)`	Overwrites only the partition for `key`	DataFrame, table, key	row count
`is_duplicate_content(log_table, content_hash)`	Whether this hash is already a distinct version, using the pulls log (ADR-0003)	log table, hash	boolean
`append_pull_log(log_table, key, load_time)`	Overwrites the pulls-log row for this pull time (pull time, content hash, load time)	log table, key, time	none
`read_as_of(table, log_table, pull_time)`	Looks up the pull's hash in the pulls log and reads the matching distinct version	tables, pull time	DataFrame
`enforce_job_limits(df, max_rows, max_runtime)`	Stops before publishing when a limit is exceeded and opens an incident	DataFrame, limits (values: OQ-6)	pass or raises `JobLimitExceeded`

C3. Publish gate (Great Expectations, module `asteroid_watch.checks`)

Job: stop publishing when data is stale, empty or out of range (write-audit-publish, BP rule 4 (write, audit, publish, so new data lands in staging and goes live only after the checks pass)). New data lands in staging first.

Name	Purpose	Takes	Returns
`build_suite_close_approach()`	Expectation suite for close approaches (freshness, not empty, value ranges; thresholds: OQ-7)	none	suite
`build_suite_fireball()`	Same for fireballs	none	suite
`run_publish_gate(table, suite)`	Runs the suite on staging data	table, suite	`GateResult` (passed or failed with failures)

Name	Purpose	Takes	Returns
`promote_staging(table)`	Moves checked data to the published location	table	none

Tested in CI with small sample files (BP Testing).

C4. Gold tables (Spark job `asteroid_watch.gold`, Phase A)

Job: from the pull tables build the tables the brief and the site read, then write them as Parquet (BP 0.4 publish step). In Phase A this is done in Spark because dbt is not used by Project 2 until Phase B (BP technology table).

Outputs, all written as Parquet files to the published location: `gold_close_approach`, `gold_fireball` and `gold_prediction_change`. The brief job (C5) writes `gold_brief` to the same location (contracts/gold_brief_tables.yaml).

Name	Purpose	Takes	Returns
`build_gold_close_approach(pull_table, as_of)`	Latest known values per approach	table, time	DataFrame
`build_gold_fireball(pull_table, as_of)`	Fireball events for the site and the brief	table, time	DataFrame
`build_prediction_changes(current_pull, previous_pull)`	Which predictions changed since the previous day's pull	two pull keys	DataFrame
`publish_gold(df, name, env)`	Writes Parquet to the publish location after the gate passes	DataFrame, name, env	path

C5. Brief generator (Airflow DAG `asteroid_watch_brief`, module `asteroid_watch.brief`)

Job: write the plain-language brief from the tables, verify every number, publish files. Components as in section 6.3.

Inputs: published gold tables (read through DuckDB); prompt version from git; model settings (OQ-8). Outputs: brief files in the site bucket (document per day and level; archive index); rows for `gold_brief` (contracts/gold_brief_tables.yaml); call-log records (platform `llm_calls`).

Name	Purpose	Takes	Returns
`BriefFacts` (dataclass)	The exact facts a brief may state: the closest approaches, the day's fireballs, the changed predictions, and the list of every number the brief may contain	n/a	n/a
`build_brief_facts(conn, brief_date)`	Queries the gold tables with DuckDB and fills `BriefFacts` (fireball window: OQ-26)	DuckDB connection, date	`BriefFacts`
`BriefWriter`	Calls the model in batch with structured output, one call per reading level or one call for both (OQ-8)	facts, prompt version, level	`BriefDraft`
`BriefDraft` (dataclass)	Draft text with sections and a list of the numbers it states	n/a	n/a
`NumberVerifier.verify(draft, facts)`	Extracts every number from the text and checks each against the facts and tables (matching rules: OQ-8)	draft, facts	`VerificationResult` (ok or list of offending numbers)
`BriefBlocked` (exception)	Raised when verification fails; nothing is published	offending numbers	n/a
`BriefPublisher.publish(draft, result)`	Writes the day's brief file and refreshes the archive index only if verification passed (ADR-0012)	draft, result	file keys
`update_archive_index(env)`	Rebuilds the list of past days	env	file key

Failure handling: a blocked brief raises `BriefBlocked`, the run fails and the alert path of step 0.5 fires; the reader sees the previous published data with the late notice (ADR-0013). The late threshold is OQ-9.

C6. Serving API (FastAPI on Lambda with DuckDB, module `asteroid_watch.api`)

Job: answer data routes from the published Parquet while the session box is off, using the serving template.

Routes: see section 10 and openapi.yaml. From Phase B a history-question route runs the metrics tools (C16) under a daily cap, a token limit and abuse limits.

Name	Purpose	Takes	Returns
`create_app()`	Builds the FastAPI app and routers	settings	app
`briefs_router`	Brief routes if served by Lambda (not applicable if ADR-0012 is approved)	date	brief
`data_router`	Close approaches, fireballs, prediction history (Phase B)	filters	JSON
`duckdb_session()`	Opens DuckDB over the published Parquet in the environment's bucket	env	connection
`handler`	Lambda entry point. The serving template (a platform component), not Project 2, decides which adapter connects Lambda to FastAPI, and Project 2 uses that adapter unchanged	event	response

C7. Web app (React and TypeScript PWA, `app/asteroid-brief`)

Job: show the brief to readers. The app can be installed on a phone.

Name	Purpose	Takes	Returns
`BriefPage`	One day's brief at the chosen reading level, in the reader's local time	date	page
`ArchivePage`	List of past days	none	page
`FireballMap`	Day's fireballs on the template map component	fireball list	map
`ChangedPredictions`	List of predictions changed since yesterday	list	list view
`ColdStartNotice`	The required cold-start sentence (BP 0.4)	none	text
`LateNotice`	Shows a visible late notice with the time of the last good run when the data is late or the last run failed; never hides data (ADR-0013). Threshold: OQ-9	last good run time, now	notice
`HistoryQuestionBox`	Phase B. Public form for history questions; shows a message when the daily cap is reached (ADR-0002)	question	answer
`usePublishedAt(brief)`	Converts UTC times to the reader's time zone (rule 15)	UTC time	local string
Web app manifest (a file that names the app and its icons) and service worker (a script the browser keeps for installed apps) (`manifest.webmanifest`, `sw.ts`)	Make the app installable; offline behavior: OQ-25	n/a	n/a

Preview deploy for every pull request comes from the platform (BP 0.4).

C8. Run metrics and status (platform component used by Project 2)

Each Project 2 DAG writes freshness and row counts to the metrics table; the status page shows each pipeline's last good run; a missed freshness target or failed run sends an email (BP 0.5). From Project 1 step 1B.9 a failed run also opens one GitHub issue. See `../Platform/design-doc.md` (placeholder).

Phase B components

C9. Backfill (Airflow DAG `asteroid_watch_backfill`)

Job: load close approaches and fireballs by date range with no gaps or duplicates. Backfills run only during work sessions, never in the nightly run (BPD; BP section "How the box is run").

Name	Purpose	Takes	Returns
`plan_backfill_windows(start, end)`	Splits a range into windows (window size: OQ-3)	dates	list of windows

Name	Purpose	Takes	Returns
`run_backfill_window(window)`	Runs ingest and write for one window	window	row count
`verify_backfill(start, end)`	Checks that every day in the range is present exactly once	dates	gaps and duplicates report

C10. dbt project (`dbt/asteroid_watch`)

Job: build the three dbt layers: staging, intermediate and marts (BP rule 6, dbt layers are staging, intermediate and marts, with unique and not_null tests on keys). The models run on DuckDB in development and in CI (the automatic checks that run on every change), and on Snowflake for the marts.

Models (names proposed): `stg_close_approach_pull`, `stg_fireball_pull`, `int_close_approach_versions`, `dim_asteroid`, `fct_prediction_history` (SCD Type 2), `mart_asteroid_prediction_chart`. Contracts: contracts/prediction_history.yaml. Tests: unique and not_null on keys (rule 6).

Snowflake refresh: Airflow task `refresh_snowflake_iceberg` at the end of each run (BP 1B.3).

Lineage: OpenLineage events to Marquez; dbt docs catalog page (BP technology table).

C11. Email edition (Lambda handlers and a sender; module `asteroid_watch.email`)

Job: double opt-in subscription, one-click unsubscribe, daily send.

Name	Purpose	Takes	Returns
`SubscriberRepository`	Reads and writes the DynamoDB sign-up table	table name	repository
`SubscribeHandler`	Validates the address, stores a pending sign-up, sends the confirmation message (token: 256-bit, single use, stored as a hash, ADR-0019; expiry and limits: OQ-14)	request	response
`ConfirmHandler`	Confirms a pending sign-up from the link	token	response
`UnsubscribeHandler`	One-click unsubscribe	token	response
`SesMailer`	Thin wrapper over SES send	message	message id
`EmailEditionSender.send_daily_edition(brief_date)`	Sends the day's verified brief to confirmed subscribers (where it runs: OQ-13)	date	send report
`build_email_message(brief, level, subscriber)`	Builds the email with unsubscribe link and header	brief, level	message

C12. Subscriber lake copy (Spark job `asteroid_watch.subscribers.copy_to_lake`)

Job: nightly copy of sign-ups into the lake and Snowflake with the address replaced by a keyed-hash code; contract contracts/subscriber_lake_copy.yaml. The key is read from Parameter Store (ADR-0007).

Name	Purpose	Takes	Returns
`read_signups(env)`	Reads the DynamoDB table by scan. Export is not used, because it writes plain addresses to S3 (ADR-0017)	env	DataFrame
`hash_email(email, key)`	HMAC-SHA-256 of the normalized address, as a full hex code (ADR-0017)	address, key	code
`copy_to_lake(df)`	Writes only the code and status fields	DataFrame	row count

C13. Feed and card (module `asteroid_watch.syndication`)

Name	Purpose	Takes	Returns
`build_rss_feed(briefs)`	Builds the RSS document (items count and content: OQ-12)	list of briefs	XML
`render_card(brief)`	Produces the shareable image card (format: OQ-11)	brief	image bytes

C14. Clarity judge (module `asteroid_watch.evals`)

Name	Purpose	Takes	Returns
`ClarityJudge.score(brief)`	LLM judge clarity score (scale: OQ-18)	brief text	score
`calibrate_judge(hand_grades)`	Compares the judge's scores with the 30 hand-graded briefs and reports how closely they agree	30 graded briefs	agreement report

Uses the platform eval harness.

C15. Prediction shifts (module `asteroid_watch.shifts`)

Name	Purpose	Takes	Returns
`detect_prediction_shifts(history)`	Statistical rule flags asteroids whose predicted distance moves between pulls (rule: OQ-16)	prediction history	flags (contracts/prediction_flag.yaml)
`ShiftExplainer.explain(flag)`	The model writes the explanation of an already flagged asteroid. It runs in the nightly batch, with the number check applied (FR-31)	flag	explanation text

C16. History questions

"History questions", "metrics tools" and "MCP metrics server" name one chain: the public history-question route calls the metrics tools, and the MCP metrics server (built in Project 1 Phase B, `../Will I have signal out there/design-doc.md`, placeholder) provides them. Project 2 adds metric definitions in MetricFlow (names proposed, for example `predicted_distance_as_of`) and the allowlisted read-only SQL fallback, logged separately (BP 1B.5). Decided by John, Oct 9, 2026: history questions are public from Phase B, with a daily cap, a token limit per answer and abuse limits (ADR-0002, ADR-0009). The numbers are open (OQ-17).

Name	Purpose	Takes	Returns
`HistoryQuestionHandler`	Lambda handler for the public route; checks the daily cap and abuse limits before any model call	question	answer or refusal (429 when capped)
`enforce_history_caps(request)`	Applies the daily cap, token limit and abuse limits (values: OQ-17)	request	allow or reject
`verify_answer_numbers(answer, query_result)`	Fails an answer that states a number not in the query result (BP 1B.5)	answer, result	pass or fail

C17. Portfolio analytics (dbt models, MetricFlow, Superset, module `asteroid_watch.analytics`)

Name	Purpose	Takes	Returns
`load_usage_events(date)`	Loads events from each site that has shipped into Snowflake, in the format and from the collection point the platform defines (ADR-0010)	date	row count
`load_cloudfront_logs(date)`	Loads CloudFront logs with the IP removed before load; addresses are removed before anything is stored beyond a short delete window (ADR-0010; window length and mechanism: OQ-19)	date	row count
`strip_ip(record)`	Ensures no raw IP address leaves the loader	record	record without IP
dbt models `stg_usage_event`, `stg_cloudfront_log`, `sessions`, `lookups`, `answers`, `feedback`, `model_calls`	Canonical tables (contracts/portfolio_analytics_marts.yaml)	n/a	n/a
MetricFlow metrics `weekly_users`, `lookup_success`, `answer_accuracy`, `cost_per_answer`	Metrics with written definitions	n/a	n/a
`publish_snapshot_charts(env)`	Renders public snapshot charts to the site each night	env	file keys

Phase C components

Not designed here: Sentry impact-risk list (terms check first) and push notifications (OQ-23).

---

8. Runtime view

8.1 Nightly run, Project 2 part

1. The platform starts the session box on spot capacity at 10:00 UTC (BPD). If spot capacity is unavailable, the night is skipped and the next run catches up (BP).
2. `asteroid_watch_ingest` pulls close approaches (with `dist-max=1LD`) and fireballs, and stores each response raw with its hash, load time and API version (C1).
3. `check_signature` compares the signature version with 1.5 and raises an alert on a difference (ADR-0004). Whether the load stops or continues after the alert is open: OQ-5.
4. Spark `write_pull` writes each pull to its own partition keyed by pull time and content hash (C2). Job limits are enforced; a job over a row or runtime limit stops before publishing and opens an incident (BP Hard limits).
5. `run_publish_gate` runs on staging data. On failure nothing is promoted and the failure sends the alert.
6. Gold tables are built and published as Parquet (C4).
7. `asteroid_watch_brief` builds facts, writes the brief, verifies numbers, publishes files (C5). In Phase B this also writes the second reading level, RSS, card and sends the email edition (C11, C13).
8. Phase B only: Snowflake refresh (`refresh_snowflake_iceberg`), dbt marts, subscriber lake copy, analytics loads, snapshot charts.
9. Each DAG writes freshness and row counts; the status page is updated (BP 0.5).
10. Iceberg maintenance is the last task of the nightly run (BP rule 7, Iceberg tables are compacted and cleaned on a schedule, as the last task of the nightly run).
11. The session box shuts down when the pipelines finish, at the latest after 2 hours. If the run reaches the 2-hour maximum, the platform records an incident, and the next run catches up (BP).

Data flow through the layers (tables and sources as in section 9):

```
flowchart TB
    subgraph src[Sources]
        s1[JPL close-approach API]:::ext
        s2[JPL fireball API]:::ext
        s3[Sign-up table&lt;br/&gt;DynamoDB, Phase B]:::store
        s4[Usage events and&lt;br/&gt;CloudFront logs, Phase B]:::ext
    end
    raw[(Raw files&lt;br/&gt;S3 with hash, load time, API version)]:::store
    subgraph lakeg[Lake tables, Iceberg]
        l1[close_approach_pull]:::cont
        l2[fireball_pull]:::cont
        l3[subscriber_lake_copy&lt;br/&gt;keyed-hash code only]:::cont
    end
    gate{{Great Expectations&lt;br/&gt;publish gate}}:::gate
    subgraph pub[Published tables]
        g1[gold_close_approach&lt;br/&gt;gold_fireball&lt;br/&gt;gold_prediction_change&lt;br/&gt;gold_brief]:::pub
        g2[dim_asteroid&lt;br/&gt;fct_prediction_history&lt;br/&gt;prediction_flag&lt;br/&gt;Phase B]:::pub
        g3[sessions, lookups, answers&lt;br/&gt;feedback, model_calls&lt;br/&gt;Phase B]:::pub
    end
    serve[Serving: Lambda API,&lt;br/&gt;brief files, PWA site]:::site
    s1 --> raw
    s2 --> raw
    raw -->|Spark write_pull| l1
    raw -->|Spark write_pull| l2
    s3 -->|hash_email| l3
    l1 --> gate
    l2 --> gate
    gate -->|gold build| g1
    l1 -->|dbt| g2
    s4 -->|dbt, no raw IP| g3
    g1 --> serve
    g2 -.->|history routes| serve
    g3 -.->|snapshot charts| serve
    classDef ext fill:#e5e7eb,stroke:#6b7280,color:#111827
```

```
    classDef cont fill:#dcfce7,stroke:#15803d,color:#0f172a
    classDef store fill:#fef3c7,stroke:#b45309,color:#0f172a
    classDef gate fill:#fee2e2,stroke:#b91c1c,color:#0f172a
    classDef pub fill:#dbeafe,stroke:#1d4ed8,color:#0f172a
    classDef site fill:#ede9fe,stroke:#6d28d9,color:#0f172a
```

8.2 Rerun of a pull

Rerun with the same data interval: `build_pull_key` yields the same key (pull time from the interval, not now()), `write_partition` overwrites only that partition, the row count is unchanged (Done-when 2A.2). A later pull with identical content writes only its pulls-log row and no new data partition, so it is not a new version (ADR-0003, Accepted, Decided by John, Oct 9, 2026).

8.3 Brief generation and blocking

`BriefWriter` produces a draft; `NumberVerifier.verify` returns offending numbers; on any offending number `BriefBlocked` is raised and nothing is published. The failure is a failed run (alert, incident record, GitHub issue from 1B.9). Readers keep seeing the previous brief with a visible late notice and the time of the last good run; nothing is hidden (ADR-0013).

8.4 Reader opens or installs the app

The browser loads the PWA from CloudFront. `BriefPage` reads the day's brief file (ADR-0012) and shows times in the reader's local time. If the latest brief is older than expected, `LateNotice` is shown with the time of the last good run (ADR-0013). Data routes call the Lambda API. The first call after a quiet period may take up to 10 seconds, and the cold-start notice says so. The session box can be off.

8.5 Subscribe, confirm, unsubscribe (Phase B)

1. The visitor submits an address to the subscribe route. `SubscribeHandler` validates the address, writes a pending record to DynamoDB and sends a confirmation message through SES.
2. The visitor follows the link in the confirmation message, and `ConfirmHandler` marks the record confirmed (double opt-in).
3. Each daily email contains a one-click unsubscribe link. When the subscriber uses it, `UnsubscribeHandler` marks the record unsubscribed.
4. Only verified test subscribers receive mail while the project stays in the SES sandbox (Decided by John, Oct 9, 2026). Production access, sender identity and bounce handling are decided later. Token design is set by default (ADR-0019). Token expiry and abuse limits are not decided (OQ-14), and where the daily send runs is not decided (OQ-13).
5. The three handlers write only to DynamoDB (BP rule 16, anything that runs 24/7 writes only to always-on storage).

8.6 Daily email send (Phase B)

After the brief is verified and published, `send_daily_edition` sends it to the confirmed subscribers. Location and timing: OQ-13. The sandbox limits recipients to verified addresses, and only verified test subscribers are used for now (ADR-0006).

8.7 Nightly subscriber copy (Phase B)

`copy_to_lake` reads the DynamoDB table and replaces each address with `hash_email(email, key)`, where the key comes from Parameter Store. It writes only the codes and statuses to the lake and Snowflake. The key never appears in a backup, because Airflow reads keys from Parameter Store and its backup holds none (BPD, Airflow keys).

8.8 Prediction-shift flags and history questions (Phase B)

dbt builds `fct_prediction_history`; `detect_prediction_shifts` flags shifts; `ShiftExplainer` explains each flag in batch, with the number check applied. A history question arrives at the public route, where the daily cap, token limit and abuse limits are checked first (ADR-0002). The question then goes through the MCP metrics tools. If no metric fits, a read-only SQL query from an allowed list runs and is logged separately (BP 1B.5). The platform traces the call in X-Ray and logs it (BP 1B.5).

8.9 Backfill (Phase B)

John starts the backfill during a work session. `plan_backfill_windows` and `run_backfill_window` load a date range, and rerunning any window gives the same result. `verify_backfill` reports gaps and duplicates. The backfill never runs in the nightly run.

8.10 Portfolio analytics (Phase B)

The loads run in the nightly run, and dbt builds the canonical tables. Superset shows the dashboard while the session box is on. `publish_snapshot_charts` writes fixed charts to the site, so they stay visible while the session box is off (rule 16).

---

9. Data model

Table names are proposed (OQ-22). Field lists marked pending cannot be completed until the first-run check confirms the JPL response (OQ-1). Contracts follow ODCS (contracts/).

Table	Layer and store	Phase	Contract	Notes
JPL close-approach API (source)	External	A	src_jpl_close_approach.yaml	`dist-max=1LD`; signature 1.5
JPL fireball API (source)	External	A	src_jpl_fireball.yaml	
`close_approach_pull`, `close_approach_pull_log`	Iceberg, lake	A	close_approach_pull.yaml	Data partition per distinct content hash plus a log row per pull (ADR-0003)
`fireball_pull`, `fireball_pull_log`	Iceberg, lake	A	fireball_pull.yaml	Same pattern
`gold_close_approach`, `gold_fireball`, `gold_prediction_change`, `gold_brief`	Parquet, published	A	gold_brief_tables.yaml	Read by Lambda and the brief job. `gold_fireball` is also the fireball baseline for Project 4 (BP 4B.3).
`dim_asteroid`, `fct_prediction_history`	Iceberg and Snowflake, dbt	B	prediction_history.yaml	SCD Type 2 (ADR-0008)
`prediction_flag`	Iceberg and Snowflake	B	prediction_flag.yaml	ADR-0009
Sign-up table (source)	DynamoDB	B	src_subscriber_signups.yaml	Personal data; point-in-time recovery on
`subscriber_lake_copy`	Iceberg and Snowflake	B	subscriber_lake_copy.yaml	Keyed-hash code only
Usage events (source)	Site events	B	src_usage_events.yaml	No raw IP
CloudFront logs (source)	S3	B	src_cloudfront_logs.yaml	IP must not reach warehouse
`sessions`, `lookups`, `answers`, `feedback`, `model_calls`	Snowflake marts via dbt	B	portfolio_analytics_marts.yaml	`model_calls` derives from platform `llm_calls` (shared table contract in `../Platform/contracts/`, placeholder)
`llm_calls`	Platform shared table	B	`../Platform/contracts/llm_calls.yaml` (placeholder)	Not redefined here

Entity-relationship view of the published tables

Only keys named in the contracts are shown; relationships follow shared keys and are not enforced foreign keys. Column lists are abbreviated (payload fields are pending, OQ-1).

```
erDiagram
    close_approach_pull ||--o{ gold_close_approach : "feeds"
    gold_close_approach ||--o{ gold_prediction_change : "approach_key"
    dim_asteroid ||--o{ fct_prediction_history : "asteroid_key"
```

```
    fct_prediction_history ||--o{ prediction_flag : "asteroid_key"
    close_approach_pull {
        timestamp pull_time
        string content_hash
        string record_key
        string api_signature_version
    }
    gold_close_approach {
        string approach_key PK
        timestamp as_of_pull_time
        timestamp event_time
    }
    gold_prediction_change {
        string approach_key PK
        timestamp previous_pull_time PK
        timestamp current_pull_time PK
    }
    gold_fireball {
        string fireball_key PK
        timestamp as_of_pull_time
        timestamp event_time
    }
    gold_brief {
        date brief_date PK
        string reading_level PK
        boolean numbers_verified
        timestamp published_at
    }
    dim_asteroid {
        string asteroid_key PK
    }
    fct_prediction_history {
        string asteroid_key PK
        string approach_key PK
        timestamp valid_from_pull_time PK
        timestamp valid_to_pull_time
        boolean is_current
    }
    prediction_flag {
        string asteroid_key PK
        timestamp flagged_at_pull_time PK
        timestamp from_pull_time
        string rule_version
    }
    subscriber_lake_copy {
        string subscriber_code PK
        date copy_date PK
        string status
        string reading_level
    }
```

Platform standards applied: event time and source key on every record (rule 2, carry event time and a source key on every record); raw files with source URL, fetch time and hash (rule 1); UTC timestamps (rule 15); each source contract diffed before loading (rule 5, each source has a contract, and the schema is diffed before loading).

---

10. API

Spec: openapi.yaml (OpenAPI 3). The public site uses API Gateway's HTTP type, as the build plan says for public pages (BP, One tool per job). Path names are proposed (OQ-22).

Path	Method	Phase	Purpose
`/v1/briefs/latest`	GET	A	Today's brief
`/v1/briefs`	GET	A	Archive index
`/v1/briefs/{date}`	GET	A	One day's brief (level query parameter in Phase B)
`/v1/close-approaches`	GET	A	Close approaches for the site map and lists
`/v1/fireballs`	GET	A	Fireballs for the map
`/v1/asteroids/{designation}/predictions`	GET	B	Prediction history for the per-asteroid chart

Path	Method	Phase	Purpose
`/v1/feed.xml`	GET	B	RSS feed
`/v1/history-questions`	POST	B	Public history question (daily cap, token limit, abuse limits)
`/v1/subscriptions`	POST	B	Start a subscription
`/v1/subscriptions/confirm`	GET	B	Confirm via token
`/v1/subscriptions/unsubscribe`	GET and POST	B	One-click unsubscribe

If ADR-0012 is approved, the brief GET routes are served as static files at the same logical URLs; the OpenAPI document still describes their contract.

Errors, limits and rate limiting: Project 2 uses no API keys and no per-key quotas, because the sources give none (the API Gateway type that supports keys and usage plans is used by Project 5). Limits for the subscribe route are open (OQ-14), and limits for the history-question route are open (OQ-17).

---

11. Deployment view

What	Where	Notes
Ingest, pull writing, gold build, brief generation, backfill, dbt, analytics loads	Session box (k3s, Airflow, Spark), nightly run from 10:00 UTC; backfills in work sessions	Spot capacity only; 4-hour limit per session; hard 2-hour maximum for the nightly run (BPD, BP)
Superset	Session box during sessions; snapshots published nightly	BP tools table
Raw files, lake tables, gold Parquet, site files	S3 (raw, lake, site buckets); governance-mode Object Lock (an S3 setting that blocks deletion or overwrite) for 30 days on live raw files only	BP Where each job runs, BPD defaults
Public site	CloudFront and S3	Free up to 1 TB and 10 million requests a month
API	Lambda (FastAPI and DuckDB) behind API Gateway HTTP API	About $1 per million requests; Lambda free tier
Sign-ups	DynamoDB, point-in-time recovery on, provisioned within the free tier	
Email	Amazon SES, sandbox only for now (verified test subscribers; sandbox limits: 200 messages a day and 1 a second)	ADR-0006
Keys	Parameter Store (SecureString) including the keyed-hash key	BPD
Warehouse	Snowflake, reading Iceberg through Glue, refresh at end of each run	X-Small, 60-second auto-suspend, 3 credits a month cap
Alerts and incidents	Platform status page, email, GitHub issue	BP 0.5, 1B.9

Environments: dev and live are fully separate (buckets, Glue databases, roles). Only merges to main reach live, only after the changed pipelines pass on sample data in CI (BP 0.2). The dev DynamoDB table and SES identity are separate; sandbox SES is the setting in dev and, for now, in live too. Dev data is deleted after 14 days (BPD). Everything is built by Terraform; modules for Project 2 are a task of the build (placeholder: `../Platform/design-doc.md` for module conventions).

Schedules: the nightly run starts at 10:00 UTC (BPD). A weekly smoke test pulls one small sample from each source (BP 1B.9). The time of day at which the email edition is sent is open (OQ-13).

---

12. Cross-cutting concerns

Security

- Every key in Parameter Store, none in code or GitHub; GitHub Actions reads them through OIDC (OpenID Connect, a way to sign in to AWS without stored keys) (BPD). One least-privilege IAM role per pipeline (BP 0.2). Dev roles cannot write live data.
- Least privilege for agents and the model: read-only tools and allowlists; source text treated as data (BP rule 13, least privilege for agents, and source text is treated as data, never as instructions). JPL text is data and never an instruction.
- Images scanned (Trivy) and secrets scanned (gitleaks) in CI (BP 0.1).
- Threat model: written in the Security and privacy document while building (DP); not before coding.

Privacy

See 3.5 and ADR-0007, ADR-0010. The inventory of personal data (email addresses in DynamoDB; derived codes elsewhere) and its deletion periods are in the Security and privacy document (while building); Project 2 values: OQ-15.

Observability

Each run writes freshness and row counts to the metrics table, and the status page shows them. The platform sends an email on failure and opens one GitHub issue per incident (no duplicates; the issue closes on recovery). From Phase B, OpenLineage sends lineage events to Marquez and X-Ray stores traces of agent runs. The platform logs every model call.

Cost limits

- Spark jobs: maximum rows and runtime (values: OQ-6).
- Model: cheapest model only, $15 monthly spend limit set before the first call, daily question cap and token limit per answer (BP Hard limits). Brief generation is batch and cached.
- Snowflake: 3 credits a month and a time limit on every query.
- DynamoDB: provisioned within the free tier, throttled rather than billed.
- S3: clean-up rules (dev data 14 days, temporary files 7 days, failed uploads 7 days, pipeline logs 90 days, security logs 1 year, old versions 30 days; raw and published kept) and a storage alarm.
- Everything on AWS: $40 budget as backup.

Reliability

Every task is idempotent (a rerun gives the same result). The platform handles spot interruptions, and the next run catches up after a skipped night (BP). A missed or failed day shows the old data with a visible late notice (ADR-0013); the late threshold is OQ-9.

Time

All timestamps UTC, converted on the page; the brief's calendar date definition: OQ-26.

Licensing and credit

Every source gets a terms note and credit line (BP pattern). JPL: OQ-24.

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13. Testing strategy

Principles (BP Testing, BPD Failure testing): ordinary unit tests with small sample files in CI; no staged production failures; real incidents recorded.

Level	What
Unit tests	`parse_close_approaches`, `parse_fireballs` on sample JSON; `check_signature` with a version other than 1.5; `build_pull_key`; `NumberVerifier` with a planted wrong number (must raise `BriefBlocked`); `hash_email`; `build_rss_feed`; `detect_prediction_shifts` on sample histories

Level	What
Spark tests	Run on GitHub's free CI machines on sample data before dev, then live (BPD). Idempotence test: run a pull twice and compare counts; as-of test; duplicate content test; job-limit test
Data tests	dbt tests on models (unique and not_null on keys, SCD Type 2 integrity) on DuckDB in CI and Snowflake; Great Expectations publish-gate tests on sample files; data contract schema diff before loading
Evals (Phase B)	Clarity judge calibrated on 30 hand-graded briefs; agreement and scores published; 10 history questions answered and checked by hand; metrics-tool eval that fails when an answer states a number not in the query result (BP 1B.5). Any CI threshold for clarity: OQ-18
Front end	Component tests for `BriefPage` time conversion; installability check on a phone (Done-when)
First-run checks	Phase A: `dist-max=1LD` returns 2020 QG (August 16, 2020), the named test case (ADR-0014; fallback: OQ-2). Phase B: test email from SES sandbox
Acceptance	The Done-when checks listed in requirements.md section 5
Manual	RSS validity check (validator: OQ-12); subscriber double opt-in with a test subscriber
Real incidents	Recorded per step 0.5

14. Alternatives considered

Topic	Options weighed	Where decided
Brief as installable app	App or website vs a feature of the data site	ADR-0001
Live vs batch model use	Live calls rejected; batch with number check	ADR-0002
Version storage	Overwrite, snapshots, blind append rejected; partition per pull	ADR-0003
Upstream change detection	Signature version alert plus contract diff	ADR-0004
Distance cap	`dist-max=1LD`	ADR-0005
Email edition	Amazon SES with double opt-in	ADR-0006
Personal data store	DynamoDB; PostgreSQL on the session box rejected	ADR-0007
History model	SCD Type 2 in dbt; snapshots rejected; Snowflake marts	ADR-0008
Shift detection	Statistical rule; model explains only	ADR-0009
Analytics	No raw IP addresses	ADR-0010
Reading levels and judge	Two levels; judge calibrated on 30	ADR-0011
Brief publication	Static files vs Lambda-built pages (Proposed)	ADR-0012
Late data	Old data with a visible late notice, never hidden	ADR-0013
Expected signature version	One value per API (1.5 close approaches, 1.2 fireballs), not one for both	ADR-0015
First-run test case	2020 QG, found in the live API	ADR-0014
Content hash	SHA-256 of raw bytes; MD5 and SHA-1 not used	ADR-0016
Subscriber copy	Keyed hash by scan; DynamoDB export rejected because it writes plain addresses to S3	ADR-0017
Upstream failure	Fail with an alert; a signature change keeps the raw file and stops publishing	ADR-0018
Confirmation tokens	256-bit, single use for confirmation, stored as hashes	ADR-0019

Platform-level alternatives (Trino, Flink, BigQuery, pgvector and DataHub removed; Docker Compose vs k3s; OpenTofu vs Terraform; PostgreSQL vs DynamoDB for visitor data) live in `../Platform/adrs/` (placeholder).

Minor decisions folded from ADRs

These were separate ADRs and are recorded here in full because each is a setting, a wording rule, a format or a repeat of a platform rule. The old numbers are kept so references still resolve.

Old ADR	Decision	Detail
ADR-0005	Query close approaches with dist-max=1LD	The close-approach request carries `dist-max=1LD`. Only passes closer than the Moon are ingested. The first-run check must pass before building on it. John decided on Oct 9, 2026 that the first-run check must name a known past close approach (Decided by John, Oct 9, 2026, so the check has a concrete case). The plan does not name one. The test case is 2020 QG (August 16, 2020). It was found in the live API on Oct 9, 2026 with dist 0.0000623 (the API page does not state the unit; read as AU, about 0.024 LD). The check passes when the response for that date range contains designation 2020 QG. The distance value is read from the response, not asserted here.
ADR-0013	When a nightly run is late or failed, pages show the old data with a visible late notice and never hide it	Show the old data with a visible late notice and the time of the last good run. Do not hide it. This replaces hiding expired data and applies to Project 2, including the Daily Asteroid Brief and the archive.
ADR-0014	The first-run check for dist-max=1LD names 2020 QG (August 16, 2020)	The first-run check uses 2020 QG on 2020-08-16. The check passes when the response for the date range 2020-08-01 to 2020-08-31 contains the designation `2020 QG`. The check reads the distance from the response and does not assert a value. The fallback if the case is missing is still open (OQ-2).
ADR-0015	The expected signature version is set for each JPL API: 1.5 for close approaches, 1.2 for fireballs	Each source contract pins its own expected signature version, and the ingest compares each response only with its own pinned value: Close approaches: 1.5. Fireballs: 1.2. The version is read from `signature.version` in both APIs. A value that differs from the pinned one raises an alert (ADR-0004). What the ingest does after the alert is set in ADR-0018.
ADR-0016	The content hash is SHA-256 of the raw response bytes	The content hash is the SHA-256 digest of the response body exactly as received, before any parsing, written as 64 lowercase hex characters. It is stored in the raw file's metadata (with source URL, load time and API version) and in the pulls log.
ADR-0017	The nightly subscriber copy reads DynamoDB by scan and hashes each address with HMAC-SHA-256; export is not used	The nightly copy job reads the sign-up table by scan, in the Spark job or a Python step on the session box. Each address is normalized (trimmed and lowercased) and hashed in memory with HMAC-SHA-256, keyed with the secret read from Parameter Store (SecureString). The code is the full 64-character lowercase hex output. It is not truncated, so codes do not collide in practice. Only the code and the status fields are written to the lake and Snowflake. No plain address is written to any file, bucket or table outside DynamoDB.
ADR-0019	Subscription confirmation and unsubscribe tokens are 256-bit random values, stored only as hashes; confirmation tokens are single use	Tokens are generated with a cryptographically secure random generator, 256 bits each, encoded as URL-safe text. DynamoDB stores the SHA-256 hash of each token, never the token itself. A leaked table then reveals no usable link. A confirmation token works once. Confirming updates the record with a condition that it is still pending, so a second use fails. Unsubscribe tokens are not single use, because unsubscribing twice must do nothing. They do not expire, so a link in an old email still works. This keeps the one-click unsubscribe rule (QR-18).

15. Risks

Risk	Handling
JPL changes the API or response format	Signature-version alert (ADR-0004), data contract diff before loading, weekly smoke test (BP 1B.9). Stop-or-continue: OQ-5
A wrong number reaches the brief	Number verification blocks publishing (ADR-0002); unit test; judge and hand checks
No brief on a day (blocked brief, spot unavailable, 2-hour maximum)	Alert, incident, catch-up on next run (BP); readers see the old data with a late notice (ADR-0013)
`dist-max=1LD` does not return a known approach	First-run check with 2020 QG (August 16, 2020), named in ADR-0014; fallback still open: OQ-2
Email edition reaches only verified test subscribers while in the SES sandbox	Accepted for now (ADR-0006); production access, sender identity and bounce handling decided later

Risk	Handling
Someone abuses the subscribe endpoint (mail to third parties, cost)	The limits are open: OQ-14. The plan's rule is that nothing in the plan may cause a surprise cost
Personal data leaks from DynamoDB	Keyed hash outside DynamoDB, key in Parameter Store, key not in backups, point-in-time recovery on, least-privilege roles
IP address reaches Snowflake through CloudFront logs	OQ-19; schema test and Done-when check for 2B.4
Analytics spans all five sites	Platform defines format and collection; Project 2 builds tables and reports and adds each site as it ships (ADR-0010)
Public history questions run up cost or are abused	Daily cap, token limit and abuse limits (ADR-0002); numbers open: OQ-17
Model cost	Cheapest model only, batch and cached, $15 limit; per-brief cost from call log
A Spark job grows beyond its expected size	Row and runtime limits; values are set after measurement (OQ-6)
Upstream terms (JPL, Sentry) not checked	OQ-24, OQ-23
Scope (five projects)	Every phase ends deployed; Phase C waits (BP Risks)
Platform risks relevant here (Marquez ARM build, 32 GB fit)	Tracked in the platform design doc, `../Platform/design-doc.md` (placeholder)

16. Open questions

Each question needs a spike (a short test run that answers it) or a decision by John. Each has a GitHub issue placeholder "issue: to be created". Answered questions stay in this table. The Status column gives the answer and the ADR number, so existing references still resolve. Open questions do not block coding of the parts they do not affect.

The Documentation Plan's table of questions still open lists no item for Project 2. It lists two platform items that affect Project 2 indirectly (does the ARM build of the pinned Marquez version succeed; does the full stack fit in 32 GB of memory); they are tracked in the platform design doc and are not repeated here. The questions below were added while drafting because the values cannot be taken from the sources.

#	Question	What answering it needs	Affects	Issue	Status (Oct 9, 2026)
OQ-1	Exact endpoint URLs, parameters, JSON field names and types of the JPL close-approach and fireball APIs, and where the signature version appears in a response	The first-run check run against the live APIs; the field lists in the data contracts stay pending until then	Contracts, C1, C2, the row grain in ADR-0008	issue: to be created	Partly answered: signature location `signature.version` confirmed in both APIs (ADR-0015). Field lists still open.
OQ-2	Fallback if `dist-max=1LD` does not return the test case, 2020 QG (August 16, 2020), named in ADR-0014	John's decision; run the check	2A.1	issue: to be created	Open (fallback). Test case named: 2020 QG, found in the live API (ADR-0014).
OQ-3	Date window requested per daily pull (and window size for backfill)	Decision plus a look at the API's behavior	C1, C9	issue: to be created	Open.
OQ-4	Hash algorithm for the content hash (the pulls log plus distinct-versions design is decided, ADR-0003)	Decision	C2, contracts	issue: to be created	Answered: SHA-256 of the raw bytes (ADR-0016), confirmed by John on Oct 9, 2026.
OQ-5	Behavior on a non-2xx response, a malformed body, a retry policy, and on a signature change (stop the load, or load and alert)	John's decision	C1, ADR-0004	issue: to be created	Answered: non-2xx and malformed responses fail with an alert; a signature change keeps the raw file and stops publishing (ADR-0018), confirmed by John on Oct 9, 2026. The retry count is a default that can change without a new ADR.

#	Question	What answering it needs	Affects	Issue	Status (Oct 9, 2026)
OQ-6	Data volumes (rows per pull, per year of backfill), maximum rows and runtime per Spark job, Project 2's share of the model spend limit, the Snowflake credits and the storage limit	Measure sizes first (BP, Data sizes: measure first); then set limits	Cost limits, C2, C4	issue: to be created	Open.
OQ-7	Freshness targets (maximum age of each pull table and of the brief at publish time) and the exact checks and thresholds in the Great Expectations suites	Decision after first pulls	C3, QR-6	issue: to be created	Open.
OQ-8	Which cheapest model, prompt, token limit per brief, structured-output shape, what counts as a number and how it matches table values (rounding, units, dates), what the brief says on a day with no close approach, one call or two for the two levels	Spike with sample facts	C5, ADR-0002, ADR-0011	issue: to be created	Open.
OQ-9	Threshold after which data counts as late and the notice wording (the rule itself is decided: show old data with a late notice, ADR-0013); whether a caught-up run publishes the missed day's brief	John's decision	C5, C7, QR-4	issue: to be created	Open. The rule is decided (ADR-0013); the threshold and wording are John's.
OQ-10	Whether a day's brief is a pre-rendered file or is built by Lambda on request (ADR-0012 is Proposed)	John's approval	C5, C6, C7, section 10	issue: to be created	Open. ADR-0012 is still Proposed.
OQ-11	Format, size and generator of the shareable image card	Decision	C13	issue: to be created	Open.
OQ-12	RSS feed content per item, number of items, reading level in the feed, and the validator that proves "the RSS feed validates"	Decision	C13, 2B.2	issue: to be created	Open.
OQ-13	Where and when the daily email is sent (nightly run or Lambda) and cost per month. Production access, sender identity and bounce handling are deferred (Decided by John, Oct 9, 2026) and are not open questions for now	Decision and an SES test (the Phase B first-run check)	ADR-0006, C11	issue: to be created	Partly answered: SES sandbox only, verified test subscribers (ADR-0006, decided by John, Oct 9, 2026). Send location and timing open.
OQ-14	Subscribe-endpoint abuse limits, confirmation token design and expiry, rate limits per address or per client without storing IP addresses	Decision	C11, QR-12	issue: to be created	Partly answered: token design (ADR-0019), confirmed by John on Oct 9, 2026. Expiry and rate limits open.
OQ-15	Keyed-hash algorithm and code length, key rotation, how the lake copy reads DynamoDB (export or scan), retention and deletion period of subscriber records and unsubscribed addresses	Decision	ADR-0007, C12	issue: to be created	Partly answered: HMAC-SHA-256 with a full code, read by scan, not export (ADR-0017), confirmed by John on Oct 9, 2026. Key rotation and retention open.
OQ-16	The statistical rule that flags prediction shifts (statistic, threshold, minimum pulls) and where flags and explanations are stored	A look at real history after the backfill	C15, ADR-0009	issue: to be created	Open.

#	Question	What answering it needs	Affects	Issue	Status (Oct 9, 2026)
OQ-17	Numbers for the public history questions: daily cap, token limit per answer, abuse limits; the 10 hand-checked history questions (public exposure is decided, ADR-0002)	Decision after measuring cost per answer	ADR-0002, ADR-0009, C16	issue: to be created	Partly answered: public exposure decided (ADR-0002, ADR-0009; John, Oct 9, 2026). The numbers are open.
OQ-18	Clarity scale, agreement metric and pass threshold for the judge; whether CI fails below a threshold	Decision after calibration	C14, QR-16	issue: to be created	Open.
OQ-19	Portfolio analytics: definition of a session without IP or cookie, length of the short delete window and the mechanism that removes IP addresses from logs (event format and collection point are the platform's, ADR-0010)	The platform design doc	C17, ADR-0010	issue: to be created	Partly answered: ownership and no-raw-IP rule decided (ADR-0010). Delete-window length, removal mechanism and session definition open.
OQ-20	Numeric latency targets for the API and the page (the plan only measures cold start and warm latency in step 1B.9)	Measurement per 1B.9	QR-17	issue: to be created	Open.
OQ-21	Which ODCS version to pin for the contracts (the sources say ODCS without a version; the files use v3.0.2 as a working choice)	Decision	contracts	issue: to be created	Open.
OQ-22	Approval of proposed names: tables, DAGs, modules, classes, functions, API paths, error model	John's review of this document	all	issue: to be created	Open.
OQ-23	Phase C: Sentry terms, definition of a "notably close" pass, push channel, Done-when checks	Phase C planning	F-C1, F-C2	issue: to be created	Open.
OQ-24	JPL data terms, required credit wording and whether a terms note is needed	Read JPL's terms	Section 3.4 and the page credits	issue: to be created	Open. Checked Oct 9, 2026: the JPL API page states no terms or credit line; JPL and NASA policy pages returned 404. Unverified.
OQ-25	PWA specifics: required manifest fields, icons, offline behavior	Decision	C7	issue: to be created	Open.
OQ-26	Definition of the brief's calendar date and the fireball window ("the day's fireballs"), given that fireball reports can arrive after the event, and time zone display	Decision plus API behavior	C5, C7	issue: to be created	Open.

17. Glossary

Term	Meaning
Asteroid Watch	Project 2
Close approach	An asteroid's pass near Earth. Here: passes closer than the Moon (`dist-max=1LD`)
LD, lunar distance	The average distance from Earth to the Moon. `1LD` is one lunar distance
Fireball	A bright meteor event recorded by sensors
Daily Asteroid Brief	The installable web app that publishes a plain-language brief each morning
PWA	Progressive web app: a website that can be installed on a phone
RSS	A feed format that feed readers poll for new items

Term	Meaning
Amazon SES	Amazon's email sending service; sandbox mode only allows verified recipients
Double opt-in	A subscriber must confirm the address from a message before receiving mail
Signature version	The API version reported in each JPL response; expected 1.5
Pull	One fetch of the JPL data; each pull is stored as its own Iceberg partition
Content hash	A hash of the response content, used with the pull time to key a partition
Iceberg	The table format used for lake tables on S3 with the Glue catalog
SCD Type 2	A table design that keeps every historical version of a row
Prediction history	How an asteroid's predicted pass changed across pulls
Gold table	A published table ready to be served or read by the brief job
Publish gate	A set of checks (Great Expectations) that must pass before data goes live
Write, audit, publish	Land new data in staging, check it, then publish
Keyed hash	A scrambled code computed from an address and a secret key
HMAC-SHA-256	A keyed hash method: SHA-256 combined with a secret key, so only someone with the key can recompute a code
SHA-256	A standard fingerprint of content: the same bytes always give the same 64-character value
Fail closed	On a failure, stop before publishing rather than publish data that may be wrong
Parameter Store	AWS Systems Manager Parameter Store, the one location for keys
MetricFlow	dbt's metric definition layer
MCP	Model Context Protocol; the metrics server exposes metrics to agents as tools
LLM judge	A model that scores another model's output against a rubric
Nightly run	The 10:00 UTC run of every pipeline on the spot session box
Session box	The spot EC2 machine running k3s, Airflow and Spark
Idempotent	Rerunning has the same effect as running once
ODCS	Open Data Contract Standard
OpenAPI	The standard format for describing HTTP APIs
DAG	Directed acyclic graph: an Airflow job made of tasks that run in a set order
Airflow	The scheduler that runs the pipelines (DAGs) on the session box
Spark	The engine that processes large tables in parallel; used here for batch jobs
dbt	A tool that builds tables from SQL models and tests them
Mart	A final table built for reports and dashboards
Snowflake	The cloud warehouse used for marts and dashboards (Phase B)
Superset	The dashboard tool; runs only while the session box is on
Great Expectations	A tool that runs data checks (here, the publish gate)
Terraform	The tool that creates the AWS resources from code
k3s	A lightweight Kubernetes, the system that runs the session box's services
CI	Continuous integration: the automatic checks (tests and scans) that run on every change in GitHub
Spot	Spare AWS server capacity at a lower price that AWS can take back at short notice
CloudFront	The AWS service that delivers the public site files
S3	AWS file storage

Term	Meaning
DynamoDB	The always-on AWS database used for visitor data such as sign-ups
Lambda	AWS service that runs code on request, with nothing billed while idle
API Gateway	The AWS service that puts a public web address in front of Lambda
Glue catalog	The AWS list of table definitions that lets Spark, DuckDB and Snowflake read the same Iceberg tables
DuckDB	A small query engine that runs inside Lambda and on the session box
Parquet	A compact column-based file format used for published tables
Partition	One stored slice of a table, for example the rows of one pull
Pulls log	A table with one row per pull (pull time, content hash, load time)
Distinct version	A pull whose content hash has not been seen before
Late notice	The visible message on a page that says the data is late, with the time of the last good run
Metrics tools	Predefined, named calculations that agents and the history-question route call instead of writing free SQL
OpenLineage, Marquez	The lineage events and the tool that shows which table was built from which
X-Ray	The AWS service that stores traces of agent and model runs
Cold start	The slow first response of Lambda after a quiet period
Hash	A short fingerprint computed from content; the same content gives the same fingerprint
OIDC	OpenID Connect: lets GitHub Actions sign in to AWS without stored keys
FastAPI	A Python web framework used for the API routes on Lambda
PITR	Point-in-time recovery: DynamoDB can restore a table to an earlier moment
Manifest	A small file that tells a browser the name, icons and start page of an installable web app
Faithfulness	Every number in a generated text is in the tables it was written from
Object Lock	An S3 setting that blocks deletion or overwrite of a file for a set time
Platform	The shared AWS base, CI and tooling (steps 0.1 to 0.5 of the build plan) built before any project
History questions, metrics tools, MCP metrics server	One chain: the public history-question route calls the metrics tools, and the MCP metrics server provides them
ADR	Architecture decision record

Part 4: Operations

Asteroid Watch/operations.md

Operations

Status: draft. Written before the first run, so the numbers below are targets to measure, not results. Last updated: Oct 9, 2026.

The platform's monitoring, alerting and incident rules are in ../Platform/adrs/0024-status-and-incidents.md (and the matching platform document). This document says what applies to Project 2.

Pipelines and what each one must show

Pipeline (DAG or job)	Runs	Must write to the metrics table	Freshness target
`asteroid_watch_ingest`	Nightly, 10:00 UTC	Run status, rows read and written, signature version	Target not yet set (open: OQ-7)
Spark `write_pull` and gold build	Nightly, after ingest	Rows written per table, limit checks	Target not yet set (OQ-7)
`asteroid_watch_brief`	Nightly, after gold	Brief published or blocked, number-check result	Brief for today published each morning (requirement FR-15)
Backfill (Phase B)	Work sessions only	Rows per window, gaps and duplicates	None; it runs on demand
Subscriber copy, analytics loads (Phase B)	Nightly	Rows copied, status counts	Target not yet set (OQ-7)

Alerts

- A failed run or a blocked brief sends an email and opens one incident record (platform step 0.5).
- A missed freshness target sends an email (platform step 0.5).
- A signature version that differs from the pinned value raises an alert and stops publishing until John reviews it (ADR-0015 and ADR-0018 of this project).

Late data

When a run fails, is skipped (no spot capacity) or hits its 2-hour maximum, each page shows the old data with a late notice and the time of the last good run. The threshold after which data counts as late is open (OQ-9).

Runbooks

No runbook exists yet. Under the platform rule, a runbook is written for each real incident, after it happens. Nothing is written in advance, so the runbook section is empty by design.

Cost

Item	Limit	Where set
Model API	$15 a month, the cheapest model, batch and cached	Provider account before the first call (platform ADR-0014)
Live history questions (Phase B)	Daily cap, token limit per answer, abuse limits (numbers open: OQ-17)	This project's ADR-0002 and ADR-0009
Spark jobs	Maximum rows and maximum runtime per job, set from measured sizes (OQ-6)	Design doc, section 7
Snowflake (Phase B)	3 credits a month, time limit per query	Platform (ADR-0013)
Session box	Share of the nightly run (about 22 hours a month for the whole plan)	Platform (ADR-0012)
DynamoDB sign-ups (Phase B)	Free tier, provisioned, throttled rather than billed	Platform (ADR-0006)

Measured cost per run and per brief is published in the README after the first month (platform rule 14). Until then the rows above are limits, not results.

Postmortems

A blameless postmortem is written for each real incident that affects readers: a wrong number, a missed brief, or a data leak. The postmortem links to the incident record and states the fix and what changed.

Part 5: Model card

Asteroid Watch/model-card.md

Model card

Status: draft, written before any model call is made. Last updated: Oct 9, 2026. The model is not named in the sources (design doc OQ-8). This card is completed when the model is chosen and the evals run.

What the model does in this project

Use	When it runs	Who sees the output	Source
Writes the daily brief at two reading levels: standard and "explain it simply"	Batch, in the nightly run	Readers, on the brief pages	Project 2 ADR-0002, ADR-0011
Scores brief clarity (an LLM judge)	Batch, for evaluation	John, and the published clarity table	ADR-0011
Explains each prediction-shift flag	Batch, in the nightly run	Readers, next to the flag	ADR-0009
Answers public history questions ("what did we believe on a given date?")	Live, Phase B only, capped	The person who asked	ADR-0002, ADR-0009

No brief page calls the model when a reader opens it. Every number the model writes is checked against the tables before publishing; a brief with a number that is not in the tables is blocked (ADR-0002).

Intended use

- Plain-language summaries of published tables for a general reader.
- Explanations of flags that a statistical rule has already raised.
- Answers to history questions, built from a metrics tool's result. A number in an answer must appear in the query result it is based on.

Not intended for

- Advice of any kind, including about safety, travel or money.
- Any statement that the tables do not support. The model is not a source of record; the tables are.
- Decisions about a person. No personal data goes into a prompt (see below).
- Use with a different model or prompt without a new version record in git.

Model and settings

Setting	Value	Status
Model	Not named. The cheapest model available, per the platform rule (ADR-0014 of the platform)	Open (OQ-8)
Prompt	Versioned in git; the version is logged with each call	To be written during the build
Token limit per brief	Not set	Open (OQ-8)
Output shape	Structured output on every call	Decided (ADR-0002, point 5)
Calls logged	Model, prompt version, tokens, latency and cost, to the platform call log	Decided (platform ADR-0015)

Evals

These are the checks the design doc requires. None has run yet.

Eval	What it tests	Pass rule	Status
Number check (unit test)	A planted wrong number blocks the brief	The brief is blocked and the job exits with an error	Planned
Judge calibration	The judge agrees with 30 briefs graded by hand	Agreement is published with the scores; threshold open (OQ-18)	Planned
History answers	10 history questions checked by hand by John	All 10 are correct; the number check passes on every answer	Planned; the number check is decided, the list of 10 is not yet written (ADR-0009)
Latency and cost	Cost per brief and per answer, from the call log	Published monthly	Planned

The history-question route is not a model-free path. If the model is replaced, these evals run again before the change goes live.

Known failure cases

None recorded yet. Failures are added here when a real incident happens, with the prompt version and the input that caused them. The incident is recorded in the platform's incident record first (platform ADR-0024).

Cases the design already expects to handle:

- A day with no close approach under one lunar distance. What the brief says is open (OQ-8).
- A fireball report that arrives after the event. The date and window rule is open (OQ-26).
- Source text that tries to give instructions. Source text is treated as data, not as instructions (platform rule 13).

Safety and privacy

- Source text from JPL and from users is data. It is never run as an instruction, and the prompt says so.
- Live history questions come from the public. The question text is not stored in full; the call log keeps a hash and the length (platform contract llm_calls, Default).
- No IP address or email address goes into a prompt.
- Daily cap, token limit per answer and abuse limits protect the monthly spend limit of $15 (ADR-0002; numbers open: OQ-17).

Update history

Date	Change
Oct 9, 2026	First draft, before any model call

Part 6: Security and privacy

Asteroid Watch/security-and-privacy.md

Security and privacy

Status: draft, written before coding. Last updated: Oct 9, 2026. Platform-wide rules (keys, dev and live, storage clean-up, the licence register) are in ../Platform/. This document covers what applies to Project 2.

1. Personal data inventory

Data	Where	Who can read it	Kept for	Deletion period
Email address of a subscriber	DynamoDB sign-up table only (ADR-0007 of this project)	The subscription handlers; the nightly copy job reads it by scan (ADR-0017)	While subscribed	Not decided (OQ-15). Unsubscribed addresses: not decided
Keyed-hash code of the address	Lake and Snowflake (`subscriber_lake_copy`)	Analysts with warehouse access	Same as the sign-up table	Follows the sign-up row (OQ-15)
Subscription status and dates	DynamoDB, then the lake copy	As above	As above	As above
Confirmation and unsubscribe tokens	DynamoDB, as SHA-256 hashes only (ADR-0019)	The handlers	Until confirmed; unsubscribe tokens do not expire	Token expiry is open (OQ-14)
Usage events (Phase B)	Always-on raw bucket, then Snowflake	Analysts	Platform clean-up rules (raw data kept, dev data 14 days)	Platform (ADR-0019 of the platform); the delete window for IP-related data is open (OQ-19)
IP addresses from CloudFront or API Gateway logs	Logs may be written; removed before anything is stored beyond a short delete window (ADR-0010 of this project)	Not stored in the warehouse	Delete window length open (OQ-19)	Open (OQ-19)
History question text (Phase B)	Not stored in full. The call log keeps a hash and the length	Platform call log readers	Platform log rule: 90 days in raw form	Platform default

Personal data is kept in one place where possible. Email addresses exist only in DynamoDB. Everything else uses the keyed-hash code.

The deletion periods marked "open" must be set by John before the email edition goes live. That is a decision for John, not a default.

2. Threat model

Threat	Example	Control	Status
Someone floods the subscribe route	Mail sent to third parties; SES and cost	Limits on the route (OQ-14); the SES sandbox only sends to verified addresses (ADR-0006)	Limits open; sandbox in place
Someone asks the history route many questions	Model cost	Daily cap, token limit per answer, abuse limits (ADR-0002)	Numbers open (OQ-17)
A guessed confirmation link	Subscribing or unsubscribing someone else	256-bit random tokens, stored as hashes, confirmation single use (ADR-0019)	Decided by default
A leaked DynamoDB table	Email addresses exposed	Keyed-hash codes outside DynamoDB; keys in Parameter Store; point-in-time recovery on (ADR-0007, ADR-0017)	In design
A leaked key	Access to AWS or the model	Keys in Parameter Store only; gitleaks in CI; one role per pipeline (platform ADR-0007)	Platform

Threat	Example	Control	Status
A wrong number published	Harm to readers	Number check blocks the brief; late notice on old data (ADR-0002, ADR-0013)	In design
Source text as instructions	JPL or user text that tells the model what to do	Source text is treated as data (platform rule 13); the model cannot take actions, only write text	In design
IP addresses in the warehouse	Privacy breach	No raw IP reaches Snowflake; schema test that no IP column exists (ADR-0010)	In design
Dev writes live data	Accidental damage	Dev roles cannot write to live (platform ADR-0020)	Platform
Upstream format change	Wrong numbers	Signature check; fail closed (ADR-0015, ADR-0018)	In design

3. Licences and terms

Source or service	Used for	Terms checked	Obligation	Status
NASA/JPL close-approach and fireball APIs	Ingest, every night	The API page was read on Oct 9, 2026; it states no terms and no credit line. The JPL and NASA policy pages returned 404	Credit line wording to be confirmed	Unverified (OQ-24). Do not publish the credit line until checked
Amazon SES	Email edition	AWS documentation, Oct 9, 2026: sandbox limits 200 a day, 1 a second, verified recipients only	Production access request is a later decision	Verified (sandbox limits)
Model API provider	Brief writing, judge, history questions	Not named in the sources	Open (OQ-8)	Unverified
Snowflake	Marts (Phase B)	Platform decision (ADR-0005 of the platform)	Resource monitor at 3 credits	Platform

Sources used by other projects (Ookla, OpenCelliD, Global Fishing Watch, Databricks Free Edition, NOAA GFS, Copernicus) are in the platform's register, not here.

4. What this document does not decide

- Deletion periods for subscriber records (OQ-15) and token expiry (OQ-14).
- Limits for the subscribe route and the history route (OQ-14, OQ-17).
- The delete window and the removal mechanism for IP addresses in logs (OQ-19).

These are John's decisions. They are listed in the design doc, section 16.

Part 7: Architecture decision records

`Asteroid Watch/adrs/0001-daily-asteroid-brief-as-installable-web-app.md`

ADR-0001: The Daily Asteroid Brief is an installable web app (PWA) with RSS and an optional email edition

Status: Accepted (Decided by John, Oct 9, 2026, because he found the brief interesting and wanted it as an app or website; recorded in Build Plan Decisions, "Projects and scope" and "Projects 3, 5, 2 and 4"). Date: Oct 9, 2026.

Context

The Daily Asteroid Brief was an existing idea. John found it interesting and wanted it as an app or website, so it was kept and expanded into its own installable web app (BPD, Projects and scope). The plan describes it as a small installable web app with a plain-language brief each morning, a page per day and an archive (BP 2A.3), with an RSS feed and an optional email edition added in Phase B (BP 2B.2).

Decision

Build the brief as an installable web app (PWA) with a page per day and an archive in Phase A. Add RSS and an optional email edition (via Amazon SES, see ADR-0006) in Phase B. The front end is React and TypeScript on the shared serving template (BP, One tool per job: Apps and APIs).

Alternatives considered

- Keep the brief as a plain feature of the asteroid site only: not chosen; John wanted it as an app or website (BPD).
- Other delivery shapes, such as a native phone app: the sources record none. No evaluation was made.

Consequences

- The app needs the installability pieces of a PWA. Which ones are required, and whether the brief works offline, are not decided (design doc OQ-25).
- The brief is a second delivery surface next to the asteroid data pages, so it needs its own publish step and archive (ADR-0012).
- The email edition brings personal data into the project; ADR-0007 covers it.
- Platform links: serving template `../Platform/design-doc.md` (placeholder), `../Platform/adrs/` (placeholder).

`Asteroid Watch/adrs/0002-brief-written-in-batch-with-number-verification.md`

ADR-0002: The brief is written in batch and every number is verified against the tables; the only public model use is capped history questions from Phase B

Status: Accepted. Amended: Decided by John, Oct 9, 2026, so that readers can ask what was believed on a past date, history questions are public from Phase B. Original basis: build plan step 2A.3; BP Model API paragraph; BPD AI decisions.

Context

The brief is plain-language text produced by a model. The plan requires that every number is checked against the tables before publishing and that the site makes no live model calls (BP 2A.3). Generated content is made in batch and cached, only the cheapest model is used for now, behind a $15 monthly spend limit set in the API account before the first call (BP Model API; BPD AI spending, Model). Public pages must not depend on the session box, the temporary server that runs the nightly pipelines (BP Platform skeleton, 0.4).

Step 2B.3 sends history questions through the metrics tools, which use a model at question time. Decided by John, Oct 9, 2026: history questions are public from Phase B, with a daily cap, a token limit and abuse limits.

Decision

1. A batch job in the nightly run writes the brief from the tables.
2. Before publishing, every number in the generated text is checked against the tables. A brief containing a number that is not in the tables is blocked and not published.
3. No brief page makes a model call. This covers the page for each day, the archive, and the brief files they read. The data routes (close approaches, fireballs, prediction history) also make no model call; flag explanations are written in batch ahead of time. The only public route that calls a model is the history-question route (`/v1/history-questions` , Phase B), described next.
4. Amended: from Phase B, history questions ("what did we believe on a given date?") are a public feature. They run at question time through the metrics tools, with SQL as the logged fallback. They are bounded by a daily cap, a token limit per answer and abuse limits (BP Hard limits: AI model). The numbers are not yet set (design doc OQ-17). Answers that state a number not in the query result fail (BP 1B.5 harness).
5. Only the cheapest model is used for now. Prompts are versioned in git, every call uses structured output, every call is logged (model, prompt version, tokens, latency, cost) to S3 (BP 1B.5 foundations, built in Project 1 Phase B).

Alternatives considered

- Live model calls when a reader opens a brief page: rejected by the plan. The cost reason is the $15 limit and caching of batch output (BP Model API).
- Publishing without number checking: rejected; the plan makes the check a Done-when item.
- History questions as a maintainer-only tool: considered, not chosen. John chose public.

Consequences

- A rule for what counts as a number and how it matches table values (rounding, units, dates, ordinal words) is needed (design doc OQ-8).
- A blocked brief means no new brief for that day. Readers keep seeing the previous data with a visible late notice (ADR-0013).
- Which model, token limit per brief and prompt are not decided (OQ-8).
- The public history route needs its own cap and abuse limits, sized so that the $15 spend limit holds. Values: OQ-17.
- The route is a new public endpoint (see openapi.yaml, `/v1/history-questions`).

`Asteroid Watch/adrs/0003-one-iceberg-partition-per-pull.md`

ADR-0003: Each pull is written to its own Iceberg partition keyed by pull time and content hash

Status: Accepted. The pulls-log resolution below was Decided by John, Oct 9, 2026, so that the partition key and the rule on identical content both hold. Basis: build plan step 2A.2 and the build plan's engineering standards 1, 2, 3 and 8, which the Context below spells out. Date: Oct 9, 2026.

Context

The project's central question is how an asteroid's predicted pass changed over time, so every pull must be kept. The engineering standards say: keep raw files immutable with source URL, fetch time and hash (rule 1); carry event time and a source key on every record (rule 2); make tasks idempotent and atomic by overwriting a partition and using the run's data interval instead of now(), keeping history by adding one partition per pull (rule 3); keep business history in rows rather than in Iceberg snapshots, because snapshot expiry deletes old versions (rule 8).

Decision

Spark writes each pull to its own Iceberg partition (a stored slice of the table), keyed by pull time and content hash. Rerunning a pull overwrites only that partition. Identical content is not counted as a new version. History is held in partitions and rows, not in snapshots (so the nightly snapshot expiry in BP rule 7 (Iceberg tables are compacted and cleaned on a schedule, as the last task of the nightly run) cannot remove it). A query "as of a past pull" selects by the pull key, not by Iceberg time travel.

Alternatives considered

- Overwrite the table on each pull (no history): contradicts the purpose of the prediction-history feature; not stated in the sources as a considered option but excluded by 2A.2.
- Rely on Iceberg snapshots for history: rejected by BP rule 8 (snapshot expiry deletes old versions).
- Append every pull blindly: rejected by BP rule 3 (overwrite a partition, never blind-append).

Consequences

- "Pull time" must be the run's data interval, not wall-clock now(), so that a rerun maps to the same partition (BP rule 3). A later pull that returns identical content must not be a new version, although the key contains both pull time and hash; the pulls log plus distinct-versions design below resolves this (Decided by John, Oct 9, 2026). The hash algorithm is open (design doc OQ-4).
- The pull window per daily pull is not decided (OQ-3).
- The SCD Type 2 prediction-history model in dbt (ADR-0008) is built on these partitions.
- Volumes (rows per pull, partitions per year) are not known; they are measured before limits are set (OQ-6).

Resolution of the partition-key question (Accepted; Decided by John, Oct 9, 2026)

The key (pull time, content hash) and the rule "identical content is not a new version" both hold with two tables. A pulls log records every pull (pull time, content hash, load time) and is overwritten per pull time, so reruns are idempotent. A distinct-versions table holds one data partition per content hash; a pull whose hash already exists writes only its log row and no new data partition. A query "as of a past pull" looks up that pull's hash in the log and reads the matching version. Only the hash algorithm remains open (design doc OQ-4). Table names `close_approach_pull_log` and `fireball_pull_log` for the logs are proposed (OQ-22).

`Asteroid Watch/adrs/0004-pin-api-signature-version-and-alert-on-change.md`

ADR-0004: Save the API signature version with every response and alert when it changes from 1.5

Status: Accepted (build plan step 2A.1; BP rule 5 (each source has a contract, and the schema is diffed before loading)). Amended by ADR-0015 (Default, Oct 9, 2026): the expected version is set for each API, 1.5 for close approaches and 1.2 for fireballs. Date: Oct 9, 2026.

Context

The JPL APIs report a signature version in their responses. An upstream change should fail loudly and early (BP rule 5, give each source a contract and diff the schema before loading). The plan sets the expected version to 1.5 (BP 2A.1). A weekly smoke test of every source is also planned from Project 1 step 1B.9 and covers format changes (BPD Source changes: data contracts plus a weekly test pull from every source).

Decision

Store the API's signature version with every raw response. If it differs from the version pinned for that API (1.5 for close approaches, 1.2 for fireballs; ADR-0015), raise an alert (email through the platform alert path, step 0.5). The data contract for each source records the expected version.

Alternatives considered

- Detect changes only by schema diff: kept as the second line of defense (rule 5) but a version change is the cheapest early signal.
- Do nothing until a run fails: rejected by BPD "Source changes" (catch format changes before the nightly run fails).

Consequences

- The plan says only "alert when it changes from 1.5". It does not say whether the load stops or continues after the alert. That behavior is decided in ADR-0018 (fail closed), confirmed by John on Oct 9, 2026 (design doc OQ-5).
- The version field is at `signature.version` in both JPL APIs (checked Oct 9, 2026). The first-run check still confirms it (OQ-1).
- When the version changes, this ADR and the contracts' pinned version are updated by a new ADR (superseding this one) once the new version is reviewed.

`Asteroid Watch/adrs/0006-email-edition-through-amazon-ses.md`

ADR-0006: The email edition is sent through Amazon SES with double opt-in and one-click unsubscribe, in the SES sandbox for now

Status: Accepted. Amended: Decided by John, Oct 9, 2026: SES sandbox only for now. Original basis: build plan step 2B.2 and the "Brief email (Project 2)" row of Risks and open items.

Context

John wanted an optional email edition of the Daily Asteroid Brief (BPD, Projects 3, 5, 2 and 4). The plan chooses Amazon SES with double opt-in and one-click unsubscribe, and says to stay in SES's sandbox unless sign-ups open publicly (BP Risks and open items). A first-run check in Phase B sends a test email from SES's sandbox mode.

AWS's SES documentation, checked Oct 9, 2026, says that while an account is in the sandbox it can send only to verified addresses and domains (or the mailbox simulator), at most 200 messages per 24 hours and 1 message per second. The Phase B first-run check still sends a test email.

Decision

Use Amazon SES. A subscriber is added only after confirming (double opt-in). Every email offers one-click unsubscribe. Sign-ups are stored in DynamoDB (ADR-0007). Decided by John, Oct 9, 2026, because the Done-when check needs only verified test subscribers: stay in the SES sandbox for now, so only verified test subscribers receive mail. Production access, sender identity and bounce handling are decided later.

Alternatives considered

- Other mail providers: not recorded in the sources.
- No email edition: the edition is "optional", and the web app and RSS cover reading without email.
- Requesting SES production access now: deferred by John.

Consequences

- The Done-when check uses a test subscriber, which the sandbox supports. The 200-message daily limit is for the whole account, not per subscriber.
- Public sign-ups cannot receive mail until production access is decided; the subscribe page and the copy on it must say the edition is in testing (wording to write during the build).
- Where the daily send runs (nightly run on the session box, or Lambda) and the monthly cost are not decided (design doc OQ-13). Whether sending from Lambda fits BP rule 16 (anything that runs 24/7 writes only to always-on storage) is part of OQ-13.

- Confirmation token design, token expiry and rate limits on the subscribe endpoint are not decided (OQ-14).

`Asteroid Watch/adrs/0007-subscribers-in-dynamodb-keyed-hash-elsewhere.md`

ADR-0007: Sign-ups are stored in DynamoDB; everywhere else an email address is a keyed-hash code

Status: Accepted (build plan step 2B.2; BPD Data storage, security and privacy: "Email addresses" and "DynamoDB"; BPD Tools: "Visitor data"). Date: Oct 9, 2026.

Context

Email addresses are personal data. The plan wants unique subscribers countable while personal data stays in one place (BPD, Email addresses: "Personal data stays in one place while unique subscribers can still be counted"). Visitor data is stored in DynamoDB because it is always on and writable at any hour while the session box is off, and free at this traffic (BPD, Visitor data). Anything that runs 24/7 writes only to always-on storage (BP rule 16, anything that runs 24/7 writes only to always-on storage). Point-in-time recovery is on for DynamoDB (BPD, DynamoDB). Keys live in Parameter Store (BPD, Key storage).

Decision

1. Sign-ups, confirmations and unsubscribes are stored in DynamoDB.
2. The nightly copy into the lake and Snowflake replaces each email address with a scrambled code (a keyed hash). The key is stored in Parameter Store (SecureString). Real addresses never leave DynamoDB. The copy reads the table by scan and hashes each address in memory with HMAC-SHA-256 (ADR-0017).
3. DynamoDB has point-in-time recovery on and is provisioned within the free tier, so heavy traffic is throttled rather than billed (BP Hard limits).

Alternatives considered

- PostgreSQL on the session box: rejected because the session box is off most of the time (BP, Visitor data row).
- Earlier options, which the sources do not name, could not take visitor writes at any hour (BPD, Visitor data).
- Storing plain addresses in the lake or Snowflake: excluded by the decision.

Consequences

- The algorithm and code length are set by ADR-0017 (default). Key rotation and what happens to old codes on rotation are not decided (design doc OQ-15).
- The retention period and deletion period of unsubscribed addresses are not decided (OQ-15). The Security and privacy document, written while building, must list the deletion period (DP).
- Because only codes leave DynamoDB, the unsubscribe flow must act on DynamoDB, not on the lake.
- Table keys and attributes of the sign-up table are specified in contracts/src_subscriber_signups.yaml (the attribute names are proposals; see design doc OQ-22).

`Asteroid Watch/adrs/0008-scd-type-2-prediction-history-with-snowflake-marts.md`

ADR-0008: Prediction history is an SCD Type 2 dbt model, with marts on Snowflake

Status: Accepted (build plan step 2B.1; BP rule 6 (dbt layers are staging, intermediate and marts, with unique and not_null tests on keys) and BP rule 8 (keep business history in rows, not in Iceberg snapshots); BPD Tools: Warehouse, Transformations). Date: Oct 9, 2026.

Context

"How each asteroid's predicted pass has changed over time" needs business history held in rows. BP rule 8 says to keep business history in rows (SCD Type 2, one partition per pull), not in snapshots. dbt is the transformation tool, with DuckDB for development and CI and Snowflake for production marts so the same models run in both places (BP, One tool per job). Snowflake reads Iceberg in place through the Glue catalog, with a refresh step at the end of each pipeline run (BP 1B.3, Snowflake setup). dbt layers are staging, intermediate and marts, with `source()` only in staging and unique and not_null tests on keys (rule 6).

Decision

In Phase B, dbt builds an SCD Type 2 prediction-history model from the per-pull Iceberg partitions (ADR-0003) and a per-asteroid chart dataset. Marts also run on Snowflake. Phase A has no dbt for Project 2 (BP technology table: dbt, Project 2: B); Phase A gold tables are produced by Spark.

Alternatives considered

- Trino, BigQuery and Databricks as dbt targets (the engines dbt could build on): the plan removed them as duplicates (BP, One tool per job).
- Iceberg snapshots as history: rejected by rule 8.

Consequences

- Phase A gold tables built in Spark must be reproducible by dbt models in Phase B with the same output; moving them is part of 2B.1.
- The grain of the SCD Type 2 rows (what one row stands for: an asteroid and approach date, an approach, or a pull) is not decided, because the JPL record fields are not yet confirmed (design doc OQ-1 and the contract note in contracts/prediction_history.yaml).
- Snowflake is capped at 3 credits a month with a time limit on every query (BPD, Snowflake; BP Hard limits).
- Platform links: `../Platform/adrs/` (warehouse, Snowflake resource monitor; placeholders).

`Asteroid Watch/adrs/0009-prediction-shift-flags-and-history-questions.md`

ADR-0009: A statistical rule flags prediction shifts, the model explains them, and history questions are public and go through the metrics tools

Status: Accepted. Amended: Decided by John, Oct 9, 2026, so that readers can ask what was believed on a past date: history questions are public from Phase B. Original basis: build plan step 2B.3; BPD AI decisions: AI placement, Agent queries, Model.

Context

2B.3: "A statistical rule flags asteroids whose predicted distance moves between pulls, and the model explains each flag. History questions ('what did we believe on a given date?') go through the metrics tools, with SQL as the logged fallback." Agent queries use defined metrics first, with raw SQL as a logged fallback, to give fewer wrong numbers (BPD). Raw SQL is read-only and allowlisted and logged separately (BP 1B.5). Least privilege and source text treated as data apply (BP rule 13, least privilege for agents, and source text is treated as data, never as instructions). Only the cheapest model is used for now (BPD, Model).

Decision

1. Detection is a statistical rule, not a model. The model only writes the explanation of an already flagged asteroid.

2. History questions are answered through the metrics tools, which the MCP metrics server built in Project 1 Phase B (BP 1B.5) exposes. These three names describe one chain: the public history-question route calls the metrics tools, and the MCP metrics server provides them. MCP (Model Context Protocol) is a standard way to give agents named tools; the metrics tools are predefined calculations. Raw SQL is a read-only, allowlisted fallback and is logged separately.
3. Decided by John, Oct 9, 2026: history questions are public, with a daily cap, a token limit per answer and abuse limits. ADR-0002 is amended to allow this from Phase B. The numbers are open (design doc OQ-17).
4. The eval harness fails when an answer states a number not in the query result (BP 1B.5 criterion), applied to history answers.

Alternatives considered

- A trained model to detect shifts: not recorded in the sources. The plan says "statistical rule".
- Free SQL generation as the first path: rejected by BPD Agent queries (defined metrics first).
- Maintainer-only history questions: not chosen.

Consequences

- The rule itself (statistic, threshold, minimum number of pulls) is unknown and needs data (OQ-16).
- A public route needs the public-endpoint protections: daily cap, token limit per answer, abuse limits, cost under the $15 limit (BP Hard limits). Values: OQ-17.
- The set of 10 history questions for the Done-when check is not yet written (OQ-17).
- Explanations of flags can be generated in batch in the nightly run and cached, consistent with ADR-0002. This is a design choice of this document, not recorded in the plan.
- Platform and Project 1 links: metrics tools and eval harness `../Will I have signal out there/design-doc.md` (placeholder), `../Platform/design-doc.md` (placeholder).

`Asteroid Watch/adrs/0010-portfolio-analytics-without-raw-ip-addresses.md`

ADR-0010: Portfolio analytics loads usage events and logs into Snowflake with no raw IP addresses; the platform defines the event format and collection

Status: Accepted. Amended: Decided by John, Oct 9, 2026, so that one place defines the event format for all five sites: ownership is split with the platform. Original basis: build plan step 2B.4; BPD Project 1 "Visitor identity"; BP rule 14 (measure and publish the numbers).

Context

2B.4: "Usage events from all five sites (no raw IP addresses) and CloudFront logs load into Snowflake. dbt builds canonical tables (sessions, lookups, answers, feedback, model calls), MetricFlow defines weekly users, lookup success, answer accuracy and cost per answer, each mart gets a written freshness target, and Superset shows the dashboard, with snapshot charts published to the site each night." Measuring and publishing the numbers is engineering standard 14. Project 1 chose a random ID on the phone and no internet addresses as the most private option (BPD, Visitor identity). Superset runs only while the session box is on; the nightly run publishes snapshot charts (BP tools table; rule 16 (anything that runs 24/7 writes only to always-on storage)).

Decision

1. Decided by John, Oct 9, 2026: the platform defines the usage-event format and the collection point. Project 2 builds the Snowflake tables and reports and adds each site as it ships.
2. Usage events carry no raw IP address. The warehouse never receives a raw IP address, including from CloudFront logs.

3. Decided by John, Oct 9, 2026 (for logs generally): logs may be written, but addresses are removed before anything is stored beyond a short delete window.
4. The canonical tables are sessions, lookups, answers, feedback and model calls. MetricFlow defines weekly users, lookup success, answer accuracy and cost per answer, and every metric has a written definition. Each mart gets a written freshness target.

Alternatives considered

- Collect IP addresses for unique-user counts: excluded by the requirement.
- Project 2 owning the event format and collection: not chosen by John.
- Another BI tool: Superset is the chosen tool (BP, One tool per job).

Consequences

- The event schema and collection endpoint are written in the platform design doc (`../Platform/design-doc.md`, placeholder). Project 2's loaders and `src_usage_events.yaml` follow it.
- Sites are added as they ship. Project 4 is built after Project 2, so its events are added later; "answer accuracy" and "cost per answer" start with the sites that have answers.
- The length of the short delete window, the mechanism that removes the IP, and the definition of a session without IP or cookie remain open (design doc OQ-19).

`Asteroid Watch/adrs/0011-two-reading-levels-and-llm-judge.md`

ADR-0011: The brief has two reading levels, and an LLM judge calibrated on 30 hand-graded briefs scores clarity

Status: Accepted (build plan step 2B.2; BP rule 13 (least privilege for agents, and source text is treated as data, never as instructions) and BP rule 14 (measure and publish the numbers); BPD AI decisions). Date: Oct 9, 2026.

Context

2B.2: "Two reading levels (standard and 'explain it simply') ... An LLM judge calibrated on 30 hand-graded briefs scores clarity." Done when: "clarity scores are published". Projects 1 and 5 calibrate their judges on 50 hand-graded answers. This project uses 30 hand-graded briefs because the plan says 30. Eval harness and judge infrastructure come from Project 1 Phase B (BP 1B.5, 1B.7).

Decision

1. Each brief is produced at two reading levels: standard and "explain it simply". Both pass the same number check (ADR-0002).
2. An LLM judge scores clarity. It is calibrated on 30 briefs graded by hand. Agreement between the judge and the hand grades is published together with the clarity scores.

Alternatives considered

- Hand grading only: it does not scale beyond the 30 calibration briefs, and the plan chooses an LLM judge.
- Calibration on 50 as in other projects: not chosen; the plan says 30 for this project.

Consequences

- The clarity scale, the agreement metric (for example correlation or exact match) and a pass threshold are not decided (design doc OQ-18). No CI failure threshold exists for clarity in the plan, unlike other projects; CI behavior is an open item.
- Whether the two levels are generated by one call or two is not decided (OQ-8).
- The "explain it simply" text must satisfy the same rule that every number is in the tables.

Asteroid Watch/adrs/0012-briefs-published-as-static-files.md

ADR-0012: Briefs are published as static files by the batch job; the API serves data queries

Status: Proposed (this decision is not recorded in the sources; it is the design this document proposes for John's approval). Date: Oct 9, 2026.

Context

The sources say public sites are static files plus Lambda, so everything stays up while the session box is off (BP, How the box is run). Public sites and audio clips are served by CloudFront and S3 (BP, Where each job runs). The serving template publishes gold tables as Parquet and answers through Lambda running DuckDB behind API Gateway (BP 0.4). The brief is generated in batch and brief pages make no model calls (ADR-0002). The plan does not say whether a day's brief is a pre-rendered file or a response built by Lambda from a table.

Decision (proposed)

The nightly batch job, after the number check passes, writes each day's brief as a file in the site bucket (one document per day and reading level) plus an updated archive index and, in Phase B, the RSS feed. The page and the PWA read those files through CloudFront. Lambda with DuckDB answers data questions (for example the per-asteroid prediction history) over published Parquet. The API in openapi.yaml lists both kinds of path.

Alternatives considered

- A Lambda endpoint that builds each brief page from the gold tables on request: it also fits the serving template, but it adds cold-start delay (the slow first response after a quiet period) to the first view and uses Lambda time. This proposal does not choose it.
- A mix: files for the brief and Lambda for history. This is the proposal.

Consequences

- If approved, "today's brief publishes on its own" means a file write at the end of the nightly run, and the late notice (ADR-0013) is a page rule based on the file's date and the last good run time.
- If rejected, the "Briefs" paths in openapi.yaml change to Lambda-served routes; only design doc sections 6, 8, 10 and 11 depend on this choice.
- Needs John's approval before coding of 2A.3 (DP: John approves each document before coding begins).

Asteroid Watch/adrs/0018-upstream-failures-fail-closed.md

ADR-0018: A failed or malformed upstream response stops the pull with an alert; a signature change keeps the raw file and stops publishing until John reviews

Status: Accepted. Confirmed by John, Oct 9, 2026. Proposed as a default by Claude under the security-first rule. Answers part of OQ-5. Date: Oct 9, 2026.

Context

ADR-0004 says to alert when the signature version changes but does not say whether the load stops. OQ-5 also asks what happens on a non-2xx response, a malformed body and a retry policy. The brief must never show a wrong number (ADR-0002), and the late-data rule shows the last good data with a notice (ADR-0013). A changed upstream format is the case most likely to produce wrong numbers without an obvious error.

Decision

1. **Non-2xx, non-JSON, or a body missing the signature field:** the ingest task fails after a small number of retries with backoff (three attempts is the default). Nothing is written to the Iceberg tables. An alert is raised.
2. **Signature version differs from the pinned value (ADR-0015):** the raw response is still stored, because raw files are kept immutable and complete (BP rule 1). The run then stops before `write_pull`, gold build and publish. The alert names the API and both versions. The site keeps the last good data with a late notice (ADR-0013).
3. Resuming needs John's review: he either accepts the new version in a new ADR that supersedes ADR-0015, or fixes the parser.

Alternatives considered

- Load and alert, publish anyway: rejected for now. A changed format could change the meaning of the fields, so publishing could show wrong numbers.
- Stop without storing the raw file: rejected. It would lose the evidence needed to diagnose the change.

Consequences

- A real upstream change holds back new briefs until John acts. That is the intended trade-off.
- The retry count is a default; it can be changed without a new ADR.
- OQ-5 is answered: John confirmed this default on Oct 9, 2026.
- Links: ADR-0004, ADR-0015; design doc C1 and 8.1 step 3.

Part 8: Data contracts

Asteroid Watch/contracts/close_approach_pull.yaml

```
# ODCS data contract: PUBLISHED lake table (Iceberg, Glue catalog).
apiVersion: v3.0.2
kind: DataContract
id: asteroid-watch.lake.close-approach-pull
name: close_approach_pull
version: 0.1.0
status: draft
domain: asteroid-watch
dataProduct: asteroid-watch-ingest
tenant: portfolio
description:
  purpose: Every pull of the JPL close-approach data, one Iceberg partition per pull keyed by pull time and content hash
(ADR-0003).
  limitations: Payload columns are pending the first-run check (OQ-1). Table name is proposed (OQ-22).
  usage: Input to the gold tables (Phase A, Spark) and to the dbt SCD Type 2 prediction-history model (Phase B).
team:
  - username: John
    role: owner
servers:
  - server: lake
    type: iceberg
    environment: prod
    location: platform-defined lake bucket and Glue database (see ../Platform/ placeholder)
  - server: lake-dev
    type: iceberg
    environment: dev
    location: separate dev bucket and Glue database (build plan 0.2)
schema:
  - name: close_approach_pull
    physicalName: close_approach_pull
    physicalType: table
    description: One row per close-approach record per pull.
    properties:
      - name: pull_time
        logicalType: timestamp
        physicalType: timestamp (UTC)
        required: true
        primaryKey: true
        primaryKeyPosition: 1
        description: The run's data interval start in UTC, not wall-clock now() (build plan rule 3, tasks are idempotent
and atomic, so overwrite one partition, use the run's data interval, and never blind-append). Part of the partition key.
      - name: content_hash
        logicalType: string
        required: true
        primaryKey: true
        primaryKeyPosition: 2
        description: Hash of the raw response content. Part of the partition key. SHA-256 of the raw bytes (ADR-0016).
Identifies the distinct version.
      - name: record_key
        logicalType: string
        required: true
        primaryKey: true
        primaryKeyPosition: 3
        description: Source key of the close-approach record within a pull (build plan rule 2, carry event time and a
source key on every record). Composition undecided until fields are confirmed (OQ-1).
      - name: source_url
        logicalType: string
        required: true
        description: URL of the request, with parameters including dist-max=1LD.
      - name: api_signature_version
        logicalType: string
        required: true
        description: Signature version saved with the response (ADR-0004).
      - name: load_time
        logicalType: timestamp
        required: true
        description: When the raw response was loaded, in UTC.
      - name: event_time
        logicalType: timestamp
        required: true
        description: The approach's event time in UTC (build plan rule 2). Source field to be confirmed (OQ-1).
      - name: payload_columns
        logicalType: object
```

```
        required: false
        description: Typed JPL record fields. List PENDING (OQ-1). Replace this entry with the typed columns after the
first-run check.
        customProperties:
          - property: fieldListStatus
            value: pending-first-run-check
  - name: close_approach_pull_log
    physicalName: close_approach_pull_log
    physicalType: table
    description: One row per pull (ADR-0003, Accepted). Overwritten per pull time, so a rerun leaves the row count
unchanged. Table name proposed (OQ-22).
    properties:
      - name: pull_time
        logicalType: timestamp
        required: true
        primaryKey: true
        primaryKeyPosition: 1
        description: Run's data interval start in UTC.
      - name: content_hash
        logicalType: string
        required: true
        description: Hash of the raw response; points to the distinct version in close_approach_pull.
      - name: load_time
        logicalType: timestamp
        required: true
        description: UTC time the raw response was loaded.
      - name: is_new_version
        logicalType: boolean
        required: true
        description: True when this pull created a new data partition, false when its hash already existed.
quality:
  - type: text
    description: Rerunning a pull leaves the row count unchanged. A query as of a past pull returns its values (2A.2).
  - type: text
    description: Passes the Great Expectations publish gate (freshness, empty results, value ranges) before promotion.
Thresholds undecided (OQ-7).
slaProperties:
  - property: frequency
    value: 1
    unit: d
customProperties:
  - property: partitionKey
    value: [pull_time, content_hash]
  - property: identicalContentRule
    value: Identical content is not counted as a new version. Decided by John, Oct 9, 2026 (ADR-0003). A pull whose
content hash already exists writes only its row in close_approach_pull_log and no new data partition.
  - property: retention
    value: Raw and published data are kept (build plan clean-up rules).
  - property: icebergMaintenance
    value: Compact, expire snapshots and remove orphan files as the last task of the nightly run (rule 7, Iceberg tables
are compacted and cleaned on a schedule, as the last task of the nightly run). History lives in partitions, not snapshots
(rule 8, keep business history in rows, not in Iceberg snapshots).
```

Asteroid Watch/contracts/fireball_pull.yaml

```
# ODCS data contract: PUBLISHED lake table (Iceberg, Glue catalog).
apiVersion: v3.0.2
kind: DataContract
id: asteroid-watch.lake.fireball-pull
name: fireball_pull
version: 0.1.0
status: draft
domain: asteroid-watch
dataProduct: asteroid-watch-ingest
tenant: portfolio
description:
  purpose: Every pull of the JPL fireball data, one Iceberg partition per pull keyed by pull time and content hash (ADR-
0003).
  limitations: Payload columns pending the first-run check (OQ-1). Table name proposed (OQ-22).
  usage: Input to gold_fireball (the day's fireballs on the map; also the baseline for Project 4 step 4B.3).
team:
  - username: John
    role: owner
servers:
  - server: lake
    type: iceberg
    environment: prod
```

```
    location: platform-defined lake bucket and Glue database (see ../Platform/ placeholder)
  - server: lake-dev
    type: iceberg
    environment: dev
schema:
  - name: fireball_pull
    physicalName: fireball_pull
    physicalType: table
    description: One row per fireball record per pull.
    properties:
      - name: pull_time
        logicalType: timestamp
        required: true
        primaryKey: true
        primaryKeyPosition: 1
        description: Run's data interval start in UTC (rule 3, tasks are idempotent and atomic, so overwrite one
partition, use the run's data interval, and never blind-append).
      - name: content_hash
        logicalType: string
        required: true
        primaryKey: true
        primaryKeyPosition: 2
        description: Hash of the raw response content (SHA-256 of the raw bytes, ADR-0016).
      - name: record_key
        logicalType: string
        required: true
        primaryKey: true
        primaryKeyPosition: 3
        description: Source key of the fireball record (rule 2, carry event time and a source key on every record).
Composition undecided (OQ-1).
      - name: source_url
        logicalType: string
        required: true
      - name: api_signature_version
        logicalType: string
        required: true
      - name: load_time
        logicalType: timestamp
        required: true
      - name: event_time
        logicalType: timestamp
        required: true
        description: Fireball event time in UTC. Source field to be confirmed (OQ-1).
      - name: payload_columns
        logicalType: object
        required: false
        description: Typed JPL fields. List PENDING (OQ-1).
        customProperties:
          - property: fieldListStatus
            value: pending-first-run-check
  - name: fireball_pull_log
    physicalName: fireball_pull_log
    physicalType: table
    description: One row per pull (ADR-0003, Accepted). Table name proposed (OQ-22).
    properties:
      - name: pull_time
        logicalType: timestamp
        required: true
        primaryKey: true
        primaryKeyPosition: 1
      - name: content_hash
        logicalType: string
        required: true
      - name: load_time
        logicalType: timestamp
        required: true
      - name: is_new_version
        logicalType: boolean
        required: true
quality:
  - type: text
    description: Rerun leaves the row count unchanged; an as-of query looks up the pull's hash in the log and returns that
version's values. A pull with an already-seen hash writes no new data partition (Decided by John, Oct 9, 2026).
slaProperties:
  - property: frequency
    value: 1
    unit: d
customProperties:
```

```
  - property: partitionKey
    value: [pull_time, content_hash]
  - property: consumers
    value: [asteroid-watch gold tables, project4 fireball baseline (build plan 4B.3)]
```

Asteroid Watch/contracts/gold_brief_tables.yaml

```
# ODCS data contract: PUBLISHED gold tables (Parquet, read by Lambda DuckDB and by the brief job).
# Column lists marked pending depend on the JPL fields (design-doc OQ-1). Table and column names are proposed (OQ-22).
apiVersion: v3.0.2
kind: DataContract
id: asteroid-watch.gold.brief-tables
name: Asteroid Watch gold tables for the Daily Asteroid Brief
version: 0.1.0
status: draft
domain: asteroid-watch
dataProduct: asteroid-watch-gold
tenant: portfolio
description:
  purpose: Published tables the brief job and the site read. Built in Spark in Phase A (dbt is not used by Project 2 until
Phase B); later reproduced by dbt models with the same output (ADR-0008).
  limitations: gold_fireball is also the fireball baseline read by Project 4 step 4B.3, so its schema changes need a
version bump and a note to Project 4.
  usage: Brief generation facts (ADR-0002), API data routes, the map. When a run is late, the page shows these tables' old
data with a late notice (ADR-0013).
team:
  - username: John
    role: owner
servers:
  - server: gold
    type: s3
    format: parquet
    environment: prod
    location: platform-defined published Parquet location (see ../Platform/ placeholder)
  - server: gold-dev
    type: s3
    format: parquet
    environment: dev
schema:
  - name: gold_close_approach
    physicalType: table
    description: Latest known values per close approach as of a pull.
    properties:
      - name: approach_key
        logicalType: string
        required: true
        primaryKey: true
        primaryKeyPosition: 1
        description: Source key of the approach (composition OQ-1).
      - name: as_of_pull_time
        logicalType: timestamp
        required: true
        description: UTC pull time the values come from.
      - name: event_time
        logicalType: timestamp
        required: true
        description: UTC time of the approach.
      - name: payload_columns
        logicalType: object
        required: false
        description: Typed approach fields. PENDING (OQ-1).
        customProperties:
          - property: fieldListStatus
            value: pending-first-run-check
  - name: gold_fireball
    physicalType: table
    description: Fireball events for the map, the brief and the Project 4 baseline.
    properties:
      - name: fireball_key
        logicalType: string
        required: true
        primaryKey: true
        primaryKeyPosition: 1
      - name: as_of_pull_time
        logicalType: timestamp
        required: true
      - name: event_time
```

```
        logicalType: timestamp
        required: true
        description: UTC time of the fireball.
      - name: payload_columns
        logicalType: object
        required: false
        description: Typed fireball fields including position and energy as the API provides them. PENDING (OQ-1).
        customProperties:
          - property: fieldListStatus
            value: pending-first-run-check
  - name: gold_prediction_change
    physicalType: table
    description: Predictions that changed between the previous and the current pull ("which predictions changed since
yesterday").
    properties:
      - name: approach_key
        logicalType: string
        required: true
        primaryKey: true
        primaryKeyPosition: 1
      - name: previous_pull_time
        logicalType: timestamp
        required: true
        primaryKey: true
        primaryKeyPosition: 2
      - name: current_pull_time
        logicalType: timestamp
        required: true
        primaryKey: true
        primaryKeyPosition: 3
      - name: changed_values
        logicalType: object
        required: true
        description: Previous and current values of the changed fields. Which fields count as a change is PENDING (OQ-1,
OQ-16).
  - name: gold_brief
    physicalType: table
    description: One row per published brief per reading level.
    properties:
      - name: brief_date
        logicalType: date
        required: true
        primaryKey: true
        primaryKeyPosition: 1
        description: Calendar date of the brief. Definition (UTC date or other) is OQ-26.
      - name: reading_level
        logicalType: string
        required: true
        primaryKey: true
        primaryKeyPosition: 2
        description: standard or explain-it-simply. "explain-it-simply" exists from Phase B (2B.2); Phase A has standard
only.
        quality:
          - type: text
            description: Value is one of standard or simple (value labels proposed, OQ-22).
      - name: text
        logicalType: string
        required: true
        description: The plain-language brief.
      - name: prompt_version
        logicalType: string
        required: true
        description: Prompt version from git (build plan 1B.5).
      - name: numbers_verified
        logicalType: boolean
        required: true
        description: True only when every number was found in the tables. A brief with false is never published (2A.3).
      - name: published_at
        logicalType: timestamp
        required: true
        description: UTC.
quality:
  - type: text
    description: Publish gate (Great Expectations) checks freshness, empty results and value ranges before publishing.
Thresholds undecided (OQ-7).
  - type: text
    description: Every number in gold_brief.text is present in the tables (2A.3).
slaProperties:
```

```
  - property: frequency
    value: 1
    unit: d
  - property: freshness
    value: null
    description: Not decided (OQ-7). The brief must be ready by morning after the 10:00 UTC nightly run (build plan
defaults).
```

Asteroid Watch/contracts/portfolio_analytics_marts.yaml

```
# ODCS data contract: PUBLISHED Snowflake canonical tables (Phase B, build plan 2B.4).
# The plan names the five tables and four metrics. Column lists are undecided because the event schema is open (design-doc
OQ-19).
apiVersion: v3.0.2
kind: DataContract
id: asteroid-watch.marts.portfolio-analytics
name: Portfolio analytics canonical tables
version: 0.1.0
status: draft
domain: portfolio-analytics
dataProduct: portfolio-analytics
tenant: portfolio
description:
  purpose: Canonical tables for weekly users, lookup success, answer accuracy and cost per answer across the five sites.
  limitations: No raw IP address. Every metric has a written definition and every mart a written freshness target (2B.4);
both are written during the build. model_calls derives from the platform shared table llm_calls
(../Platform/contracts/llm_calls.yaml, placeholder).
  usage: MetricFlow metrics, the Superset dashboard, nightly public snapshot charts.
team:
  - username: John
    role: owner
servers:
  - server: snowflake
    type: snowflake
    environment: prod
schema:
  - name: sessions
    physicalType: table
    properties:
      - name: session_key
        logicalType: string
        required: true
        primaryKey: true
        primaryKeyPosition: 1
        description: Session definition without IP or cookie undecided (OQ-19).
      - name: session_start
        logicalType: timestamp
        required: true
  - name: lookups
    physicalType: table
    properties:
      - name: lookup_key
        logicalType: string
        required: true
        primaryKey: true
        primaryKeyPosition: 1
      - name: site
        logicalType: string
        required: true
      - name: succeeded
        logicalType: boolean
        required: true
        description: Basis of the lookup_success metric; definition to be written.
  - name: answers
    physicalType: table
    properties:
      - name: answer_key
        logicalType: string
        required: true
        primaryKey: true
        primaryKeyPosition: 1
      - name: site
        logicalType: string
        required: true
      - name: accurate
        logicalType: boolean
        required: false
```

```
        description: Basis of the answer_accuracy metric; how accuracy is judged per answer is undecided (OQ-19).
  - name: feedback
    physicalType: table
    properties:
      - name: feedback_key
        logicalType: string
        required: true
        primaryKey: true
        primaryKeyPosition: 1
      - name: site
        logicalType: string
        required: true
      - name: vote
        logicalType: string
        required: true
        description: For example thumbs up or thumbs down (stored in DynamoDB for Projects 1 and 4; build plan 1B.7,
4B.3).
  - name: model_calls
    physicalType: table
    properties:
      - name: call_key
        logicalType: string
        required: true
        primaryKey: true
        primaryKeyPosition: 1
      - name: cost
        logicalType: number
        required: true
        description: Basis of cost_per_answer. Currency and unit as in llm_calls.
customProperties:
  - property: metrics
    value: [weekly_users, lookup_success, answer_accuracy, cost_per_answer]
  - property: metricNamesStatus
    value: Names proposed (OQ-22); definitions to be written during the build.
quality:
  - type: text
    description: No raw IP address reaches the warehouse (Done-when 2B.4). One metric gives the same value in Snowflake
and DuckDB (pattern from 1B.3).
slaProperties:
  - property: freshness
    value: null
    description: A written freshness target is required for each mart; values are not set (OQ-7).
```

Asteroid Watch/contracts/prediction_flag.yaml

```
# ODCS data contract: PUBLISHED table (Phase B). Flags from the statistical rule plus the model's explanation.
# The rule is undecided (design-doc OQ-16); the columns below are the minimum implied by build plan 2B.3.
apiVersion: v3.0.2
kind: DataContract
id: asteroid-watch.marts.prediction-flag
name: prediction_flag
version: 0.1.0
status: draft
domain: asteroid-watch
dataProduct: asteroid-watch-history
tenant: portfolio
description:
  purpose: Asteroids whose predicted distance moved between pulls, each with an explanation (build plan 2B.3; ADR-0009).
  limitations: Statistic, threshold and minimum pulls are undecided (OQ-16). Explanations are generated by the model in
batch and pass the number check.
  usage: The flags shown on the site; input to history questions.
team:
  - username: John
    role: owner
servers:
  - server: lake
    type: iceberg
    environment: prod
schema:
  - name: prediction_flag
    physicalType: table
    properties:
      - name: asteroid_key
        logicalType: string
        required: true
        primaryKey: true
        primaryKeyPosition: 1
```

```
      - name: flagged_at_pull_time
        logicalType: timestamp
        required: true
        primaryKey: true
        primaryKeyPosition: 2
        description: UTC pull time at which the shift was detected.
      - name: from_pull_time
        logicalType: timestamp
        required: true
        description: UTC pull time of the earlier prediction compared.
      - name: distance_before
        logicalType: number
        required: true
        description: Predicted distance before. Unit follows the source field (OQ-1).
      - name: distance_after
        logicalType: number
        required: true
      - name: rule_version
        logicalType: string
        required: true
        description: Version of the statistical rule that fired (rule itself OQ-16).
      - name: explanation
        logicalType: string
        required: true
        description: Model-written explanation. Every number in it must appear in the tables.
      - name: prompt_version
        logicalType: string
        required: true
quality:
  - type: text
    description: Flags appear with explanations; 10 history questions return answers checked by hand (2B.3).
slaProperties:
  - property: frequency
    value: 1
    unit: d
```

Asteroid Watch/contracts/prediction_history.yaml

```
# ODCS data contract: PUBLISHED dbt models (Phase B), Iceberg and Snowflake marts.
# The SCD Type 2 grain depends on confirmed JPL fields (design-doc OQ-1). Names proposed (OQ-22).
apiVersion: v3.0.2
kind: DataContract
id: asteroid-watch.marts.prediction-history
name: Asteroid prediction history (SCD Type 2)
version: 0.1.0
status: draft
domain: asteroid-watch
dataProduct: asteroid-watch-history
tenant: portfolio
description:
  purpose: Shows how each asteroid's predicted pass changed over time (build plan 2B.1; ADR-0008).
  limitations: Grain and tracked attributes pending OQ-1. Business history is kept in rows, not in Iceberg snapshots
(build plan rule 8, keep business history in rows, not in Iceberg snapshots).
  usage: Per-asteroid chart, shift flags (ADR-0009), history questions ("what did we believe on a given date?").
team:
  - username: John
    role: owner
servers:
  - server: lake
    type: iceberg
    environment: prod
  - server: snowflake
    type: snowflake
    environment: prod
    description: Snowflake reads the Iceberg tables in place through the Glue catalog; a refresh runs at the end of each
pipeline run (build plan 1B.3).
schema:
  - name: dim_asteroid
    physicalType: table
    description: One row per asteroid.
    properties:
      - name: asteroid_key
        logicalType: string
        required: true
        primaryKey: true
        primaryKeyPosition: 1
        description: Designation or other source key. PENDING (OQ-1).
```

```
      - name: asteroid_attributes
        logicalType: object
        required: false
        description: PENDING (OQ-1).
  - name: fct_prediction_history
    physicalType: table
    description: SCD Type 2 rows. One row per asteroid, approach and period during which a predicted pass held the same
values.
    properties:
      - name: asteroid_key
        logicalType: string
        required: true
        primaryKey: true
        primaryKeyPosition: 1
      - name: approach_key
        logicalType: string
        required: true
        primaryKey: true
        primaryKeyPosition: 2
      - name: valid_from_pull_time
        logicalType: timestamp
        required: true
        primaryKey: true
        primaryKeyPosition: 3
        description: Pull time (UTC) at which these values were first seen.
      - name: valid_to_pull_time
        logicalType: timestamp
        required: false
        description: Pull time (UTC) at which the values were replaced; null for the current row.
      - name: is_current
        logicalType: boolean
        required: true
      - name: predicted_values
        logicalType: object
        required: true
        description: Predicted pass values, including the predicted distance. Fields PENDING (OQ-1).
quality:
  - type: text
    description: unique and not_null tests on keys (build plan rule 6, dbt layers are staging, intermediate and marts,
with unique and not_null tests on keys). A multi-year backfill loads without gaps or duplicates (2B.1).
  - type: text
    description: The same model gives the same result on DuckDB and on Snowflake.
slaProperties:
  - property: frequency
    value: 1
    unit: d
  - property: freshness
    value: null
    description: Each mart gets a written freshness target (build plan 2B.4); value not yet set (OQ-7).
```

Asteroid Watch/contracts/src_cloudfront_logs.yaml

```
# ODCS data contract: SOURCE. CloudFront logs (Phase B, build plan 2B.4).
# Field list is PENDING. CloudFront logs contain a client address by nature (to be confirmed); the contract requires it is
removed before the warehouse (OQ-19).
apiVersion: v3.0.2
kind: DataContract
id: asteroid-watch.src.cloudfront-logs
name: CloudFront logs (source)
version: 0.1.0
status: draft
domain: portfolio-analytics
dataProduct: portfolio-analytics
tenant: portfolio
description:
  purpose: Request logs of the public sites, loaded into Snowflake for the portfolio dashboard.
  limitations: Decided by John, Oct 9, 2026 (ADR-0010). Logs may be written, but addresses are removed before anything is
stored beyond a short delete window. The window length and removal mechanism are open (OQ-19). Any client IP field must be
dropped or irreversibly altered before load.
  usage: Staging model for sessions and lookups.
team:
  - username: John
    role: owner
servers:
  - server: cloudfront-logs
    type: s3
    environment: prod
```

```
    description: Log bucket and format set up by the platform (see ../Platform/ placeholder). Security logs are kept 1
year (BPD); whether that covers CloudFront logs is not stated.
schema:
  - name: cloudfront_log_record
    physicalType: record
    properties:
      - name: request_time
        logicalType: timestamp
        required: true
        description: UTC.
      - name: uri
        logicalType: string
        required: true
      - name: status
        logicalType: integer
        required: true
      - name: other_fields
        logicalType: object
        required: false
        description: PENDING (OQ-19). Any raw client IP address is excluded.
quality:
  - type: text
    description: No raw IP address reaches Snowflake (Done-when 2B.4).
slaProperties:
  - property: freshness
    value: null
    description: Not set (OQ-7).
```

Asteroid Watch/contracts/src_jpl_close_approach.yaml

```
# ODCS data contract: SOURCE. JPL close-approach API.
# The ODCS version is a working choice (design-doc OQ-21).
# Field-level schema is PENDING the Phase A first-run check (design-doc OQ-1):
# the three source documents do not list the API's field names, so none are invented here.
apiVersion: v3.0.2
kind: DataContract
id: asteroid-watch.src.jpl-close-approach
name: JPL close-approach API (source)
version: 0.1.0
status: draft
domain: asteroid-watch
dataProduct: asteroid-watch-ingest
tenant: portfolio
description:
  purpose: Daily close approaches closer than the Moon, pulled by the Airflow ingest DAG (build plan step 2A.1).
  limitations: Request uses dist-max=1LD only (ADR-0005). The data terms and credit wording are not recorded in the
sources (design-doc OQ-24).
  usage: Raw responses are stored immutable with source URL, hash, load time and API signature version (build plan rule 1
(keep raw files immutable, with source URL, fetch time and hash); ADR-0004).
team:
  - username: John
    role: owner
servers:
  - server: jpl-close-approach
    type: api
    environment: external
    description: The JPL close-approach API. Exact base URL and path to be confirmed in the first-run check (OQ-1).
schema:
  - name: close_approach_response
    physicalType: json-document
    description: One JSON response of the close-approach API for one request.
    properties:
      - name: signature_version
        logicalType: string
        required: true
        description: API signature version reported in the response at signature.version (checked Oct 9, 2026). Expected
value 1.5. A different value raises an alert (ADR-0004, ADR-0015).
        quality:
          - type: text
            description: Must equal 1.5; any other value alerts.
      - name: records
        logicalType: array
        required: true
        description: The list of close-approach records. Record field names and types are pending (OQ-1).
        customProperties:
          - property: fieldListStatus
            value: pending-first-run-check
```

```
quality:
  - type: text
    description: Request carries dist-max=1LD. Response must be HTTP 200 JSON. Behavior on non-200 and on a version change
is open (OQ-5).
slaProperties:
  - property: frequency
    value: 1
    unit: d
    description: One pull per day in the nightly run (build plan 2A.1).
  - property: freshness
    value: null
    description: Numeric freshness target not decided (OQ-7).
customProperties:
  - property: expectedSignatureVersion
    value: "1.5"
  - property: requestParameters
    value:
      dist-max: 1LD
  - property: datePullWindow
    value: undecided (OQ-3)
  - property: weeklySmokeTest
    value: true
    description: A weekly smoke test pulls one small sample from this source (build plan step 1B.9).
```

Asteroid Watch/contracts/src_jpl_fireball.yaml

```
# ODCS data contract: SOURCE. JPL fireball API.
# Field-level schema is PENDING the Phase A first-run check (design-doc OQ-1).
apiVersion: v3.0.2
kind: DataContract
id: asteroid-watch.src.jpl-fireball
name: JPL fireball API (source)
version: 0.1.0
status: draft
domain: asteroid-watch
dataProduct: asteroid-watch-ingest
tenant: portfolio
description:
  purpose: Daily fireball events pulled by the Airflow ingest DAG (build plan step 2A.1) for the brief, the map and the
Project 4 fireball baseline (build plan 4B.3).
  limitations: Terms and credit wording not recorded in the sources (OQ-24). This API reports signature 1.2, not 1.5
(checked Oct 9, 2026; ADR-0015). How late reports arrive is not confirmed (OQ-26).
  usage: Raw responses stored immutable with source URL, hash, load time and API version.
team:
  - username: John
    role: owner
servers:
  - server: jpl-fireball
    type: api
    environment: external
    description: The JPL fireball API. Exact base URL and path to be confirmed (OQ-1).
schema:
  - name: fireball_response
    physicalType: json-document
    description: One JSON response of the fireball API for one request.
    properties:
      - name: signature_version
        logicalType: string
        required: true
        description: API signature version in the response at signature.version. Expected value 1.2 (read Oct 9, 2026;
ADR-0015). A different value raises an alert (ADR-0004).
      - name: records
        logicalType: array
        required: true
        description: The list of fireball records. Field names and types are pending (OQ-1).
        customProperties:
          - property: fieldListStatus
            value: pending-first-run-check
quality:
  - type: text
    description: Response must be HTTP 200 JSON. Behavior on non-200 is open (OQ-5).
slaProperties:
  - property: frequency
    value: 1
    unit: d
  - property: freshness
    value: null
```

```
    description: Not decided (OQ-7).
customProperties:
  - property: expectedSignatureVersion
    value: "1.2"  # ADR-0015
  - property: datePullWindow
    value: undecided (OQ-3)
  - property: weeklySmokeTest
    value: true
```

Asteroid Watch/contracts/src_subscriber_signups.yaml

```
# ODCS data contract: SOURCE. DynamoDB sign-up table for the email edition (Phase B).
# PERSONAL DATA. Real email addresses exist only in this table (ADR-0007). Attribute names proposed (OQ-22).
apiVersion: v3.0.2
kind: DataContract
id: asteroid-watch.src.subscriber-signups
name: Subscriber sign-ups (DynamoDB)
version: 0.1.0
status: draft
domain: asteroid-watch
dataProduct: asteroid-watch-email
tenant: portfolio
description:
  purpose: State of each email-edition subscription (pending, confirmed or unsubscribed; double opt-in; build plan 2B.2;
ADR-0006).
  limitations: Contains personal data. Deletion periods are not decided (OQ-15). Written only by the subscription Lambda
handlers, which write only to always-on storage (rule 16, anything that runs 24/7 writes only to always-on storage).
  usage: Sending the daily edition to verified test subscribers while in the SES sandbox (ADR-0006); nightly copy to the
lake with the address replaced by a keyed-hash code (subscriber_lake_copy.yaml).
team:
  - username: John
    role: owner
servers:
  - server: dynamodb
    type: dynamodb
    environment: prod
    description: DynamoDB with point-in-time recovery on, provisioned within the free tier so heavy traffic is throttled
rather than billed (build plan Hard limits).
  - server: dynamodb-dev
    type: dynamodb
    environment: dev
schema:
  - name: subscriber_signup
    physicalType: table
    properties:
      - name: email
        logicalType: string
        required: true
        primaryKey: true
        primaryKeyPosition: 1
        classification: pii
        description: The real address. Never leaves DynamoDB. Whether the partition key is the address or a code is
undecided (OQ-15).
      - name: status
        logicalType: string
        required: true
        description: One of pending, confirmed, unsubscribed (labels proposed).
      - name: reading_level
        logicalType: string
        required: false
        description: Which reading level the subscriber receives. Whether the email edition offers both levels is not
stated in the plan (OQ-13).
      - name: created_at
        logicalType: timestamp
        required: true
        description: UTC.
      - name: confirmed_at
        logicalType: timestamp
        required: false
        description: UTC.
      - name: unsubscribed_at
        logicalType: timestamp
        required: false
        description: UTC.
      - name: confirmation_token
        logicalType: string
        required: false
```

```
        description: Token is 256-bit, single use for confirmation, and stored as a hash (ADR-0019). Expiry undecided (OQ-
14).
quality:
  - type: text
    description: A subscriber receives mail only after confirming. Every email carries a one-click unsubscribe (2B.2).
slaProperties:
  - property: retention
    value: null
    description: Retention and deletion period undecided (OQ-15); to be listed in the Security and privacy document.
customProperties:
  - property: pointInTimeRecovery
    value: true
  - property: containsPersonalData
    value: true
```

Asteroid Watch/contracts/src_usage_events.yaml

```
# ODCS data contract: SOURCE. Usage events from all five sites (Phase B, build plan 2B.4).
# Decided by John, Oct 9, 2026 (ADR-0010): the platform defines the event format and collection point; this contract must
be aligned to the platform's definition (../Platform/ placeholder) when it exists. Project 2 loads each site as it ships.
The columns below are the minimum implied by the plan. No raw IP addresses.
apiVersion: v3.0.2
kind: DataContract
id: asteroid-watch.src.usage-events
name: Portfolio usage events (source)
version: 0.1.0
status: draft
domain: portfolio-analytics
dataProduct: portfolio-analytics
tenant: portfolio
description:
  purpose: Events from the five sites that feed sessions, lookups, answers, feedback and model calls.
  limitations: Must not contain a raw IP address. Format and collection point are defined by the platform (ADR-0010).
Sites are added as they ship; Project 4 is built after Project 2.
  usage: Loaded into Snowflake; dbt builds the canonical tables.
team:
  - username: John
    role: owner
servers:
  - server: events
    type: s3
    environment: prod
    description: Location is the platform's collection point. Anything that runs 24/7 writes only to always-on storage
(rule 16, anything that runs 24/7 writes only to always-on storage).
schema:
  - name: usage_event
    physicalType: record
    properties:
      - name: event_time
        logicalType: timestamp
        required: true
        description: UTC (rule 15, store every timestamp in UTC and convert only on the page).
      - name: site
        logicalType: string
        required: true
        description: Which of the five sites emitted the event.
      - name: event_type
        logicalType: string
        required: true
        description: Event type values come from the platform event format.
      - name: visitor_or_session_id
        logicalType: string
        required: false
        description: Identity approach for sessions without IP address or cookie is open (OQ-19). Project 1 uses a random
ID on the phone for hiker reports (BPD).
quality:
  - type: text
    description: No raw IP address in any field (Done-when 2B.4; QR-10).
slaProperties:
  - property: freshness
    value: null
    description: A written freshness target is required per mart (2B.4); not yet set (OQ-7).
```

Asteroid Watch/contracts/subscriber_lake_copy.yaml

```
# ODCS data contract: PUBLISHED lake table (Phase B). Nightly copy of sign-ups. NO email addresses.
apiVersion: v3.0.2
kind: DataContract
id: asteroid-watch.lake.subscriber-lake-copy
name: subscriber_lake_copy
version: 0.1.0
status: draft
domain: asteroid-watch
dataProduct: asteroid-watch-email
tenant: portfolio
description:
  purpose: Lets unique subscribers be counted while personal data stays in one place (BPD, Email addresses; ADR-0007).
  limitations: The code is a keyed hash whose key is in Parameter Store. HMAC-SHA-256 with a full hex code, read by scan
(ADR-0017). Key rotation is undecided (OQ-15). The key is never in an Airflow backup (BPD).
  usage: Counts and analytics in the lake and Snowflake.
team:
  - username: John
    role: owner
servers:
  - server: lake
    type: iceberg
    environment: prod
  - server: snowflake
    type: snowflake
    environment: prod
schema:
  - name: subscriber_lake_copy
    physicalType: table
    properties:
      - name: subscriber_code
        logicalType: string
        required: true
        primaryKey: true
        primaryKeyPosition: 1
        classification: pseudonymized
        description: Keyed-hash code that replaces the address.
      - name: copy_date
        logicalType: date
        required: true
        primaryKey: true
        primaryKeyPosition: 2
        description: Date (UTC) of the nightly copy; one partition per copy.
      - name: status
        logicalType: string
        required: true
        description: pending, confirmed or unsubscribed.
      - name: reading_level
        logicalType: string
        required: false
      - name: created_at
        logicalType: timestamp
        required: true
      - name: confirmed_at
        logicalType: timestamp
        required: false
      - name: unsubscribed_at
        logicalType: timestamp
        required: false
quality:
  - type: text
    description: No column may contain an email address. A search of lake and warehouse tables for the address pattern
returns nothing (QR-9).
slaProperties:
  - property: frequency
    value: 1
    unit: d
    description: Copied in the nightly run.
```

Part 9: API specification

Asteroid Watch/openapi.yaml

```
openapi: 3.0.3
info:
  title: Asteroid Watch API
  version: 0.1.0
  description: |
    Endpoints of Project 2 (Asteroid Watch and the Daily Asteroid Brief).
    Status: draft. Every path name, parameter name and schema name is proposed (design-doc OQ-22).
    Fields marked "pending" depend on the JPL response fields confirmed in the Phase A first-run check (design-doc OQ-
1).
    Public pages use API Gateway's HTTP type (build plan, One tool per job). Project 2 uses no API keys and no usage
plans (none are in the sources).
    No request rate limits are decided yet (design-doc OQ-14). The first request after a period of inactivity may take
up to 10 seconds while the service initializes.
    A late or failed nightly run does not remove data: responses carry the published date and the page shows a late
notice (ADR-0013).
    If ADR-0012 is approved, the brief GET routes are served as static files at these same paths; the contract below
still applies.
    All timestamps are UTC (build plan rule 15, store every timestamp in UTC and convert only on the page) and are
converted to local time only on the page.
servers:
  - url: https://{host}
    description: Host names for dev and live are defined by the platform (see ../Platform/ placeholder).
    variables:
      host:
        default: platform-defined
tags:
  - name: Briefs
    description: Phase A.
  - name: Data
    description: Phase A, with history in Phase B.
  - name: History questions
    description: Phase B. Public, capped (ADR-0002, ADR-0009).
  - name: Subscriptions
    description: Phase B, email edition (ADR-0006, ADR-0007).
  - name: Feed
    description: Phase B.
paths:
  /v1/briefs/latest:
    get:
      tags: [Briefs]
      operationId: getLatestBrief
      summary: Today's brief (the most recent published brief)
      description: Returns the most recent brief that passed number verification. What is returned when today's brief
is blocked or missing is undecided (design-doc OQ-9).
      parameters:
        - $ref: '#/components/parameters/Level'
      responses:
        '200':
          description: The brief.
          content:
            application/json:
              schema:
                $ref: '#/components/schemas/Brief'
        '404':
          $ref: '#/components/responses/NotFound'
        '500':
          $ref: '#/components/responses/ServerError'
  /v1/briefs:
    get:
      tags: [Briefs]
      operationId: listBriefs
      summary: Archive index
      description: Lists published days, newest first. Paging style and page size are undecided (design-doc OQ-22).
      responses:
        '200':
          description: The archive index.
          content:
            application/json:
              schema:
                $ref: '#/components/schemas/BriefIndex'
        '500':
```

```
          $ref: '#/components/responses/ServerError'
  /v1/briefs/{date}:
    get:
      tags: [Briefs]
      operationId: getBrief
      summary: One day's brief
      parameters:
        - name: date
          in: path
          required: true
          description: Brief date (YYYY-MM-DD). Definition of the date (UTC or other) is undecided (design-doc OQ-26).
          schema:
            type: string
            format: date
        - $ref: '#/components/parameters/Level'
      responses:
        '200':
          description: The brief for that day.
          content:
            application/json:
              schema:
                $ref: '#/components/schemas/Brief'
        '400':
          $ref: '#/components/responses/BadRequest'
        '404':
          $ref: '#/components/responses/NotFound'
        '500':
          $ref: '#/components/responses/ServerError'
  /v1/close-approaches:
    get:
      tags: [Data]
      operationId: listCloseApproaches
      summary: Close approaches closer than the Moon
      description: Reads the published gold_close_approach table with DuckDB. Only passes under 1 lunar distance exist
in the data (ADR-0005).
      parameters:
        - name: from
          in: query
          required: false
          description: Start date (UTC, YYYY-MM-DD). Default window undecided (design-doc OQ-3).
          schema:
            type: string
            format: date
        - name: to
          in: query
          required: false
          schema:
            type: string
            format: date
      responses:
        '200':
          description: Close approaches.
          content:
            application/json:
              schema:
                $ref: '#/components/schemas/CloseApproachList'
        '400':
          $ref: '#/components/responses/BadRequest'
        '500':
          $ref: '#/components/responses/ServerError'
  /v1/fireballs:
    get:
      tags: [Data]
      operationId: listFireballs
      summary: Fireball events
      description: Reads the published gold_fireball table. Used for the map.
      parameters:
        - name: from
          in: query
          required: false
          schema:
            type: string
            format: date
        - name: to
          in: query
          required: false
```

```
          schema:
            type: string
            format: date
      responses:
        '200':
          description: Fireballs.
          content:
            application/json:
              schema:
                $ref: '#/components/schemas/FireballList'
        '400':
          $ref: '#/components/responses/BadRequest'
        '500':
          $ref: '#/components/responses/ServerError'
  /v1/asteroids/{designation}/predictions:
    get:
      tags: [Data]
      operationId: getAsteroidPredictions
      summary: Prediction history of one asteroid (Phase B)
      description: Reads fct_prediction_history for the per-asteroid chart ("what did we believe, and when").
      parameters:
        - name: designation
          in: path
          required: true
          description: Asteroid identifier as given by the source. Exact identifier field is pending (design-doc OQ-1).
          schema:
            type: string
        - name: as_of
          in: query
          required: false
          description: Return what was believed at this UTC time ("what did we believe on a given date?").
          schema:
            type: string
            format: date-time
      responses:
        '200':
          description: Prediction history.
          content:
            application/json:
              schema:
                $ref: '#/components/schemas/PredictionHistory'
        '400':
          $ref: '#/components/responses/BadRequest'
        '404':
          $ref: '#/components/responses/NotFound'
        '500':
          $ref: '#/components/responses/ServerError'
  /v1/feed.xml:
    get:
      tags: [Feed]
      operationId: getRssFeed
      summary: RSS feed of the brief (Phase B)
      description: Must validate as RSS (Done-when 2B.2). Items, item count and reading level are undecided (design-doc
OQ-12).
      responses:
        '200':
          description: The feed.
          content:
            application/rss+xml:
              schema:
                type: string
        '500':
          $ref: '#/components/responses/ServerError'
  /v1/history-questions:
    post:
      tags: [History questions]
      operationId: askHistoryQuestion
      summary: Ask a history question (Phase B, public)
      description: |
        Answers questions such as "what did we believe on a given date?" through the metrics tools, with read-only
allowlisted SQL as the logged fallback.
        Decided by John, Oct 9, 2026: public, with a daily cap, a token limit per answer and abuse limits. The numbers
are undecided (design-doc OQ-17).
        Any number in the answer is checked against the query result.
      requestBody:
        required: true
```

```
        content:
          application/json:
            schema:
              $ref: '#/components/schemas/HistoryQuestionRequest'
      responses:
        '200':
          description: The answer, with the query result it is based on.
          content:
            application/json:
              schema:
                $ref: '#/components/schemas/HistoryAnswer'
        '400':
          $ref: '#/components/responses/BadRequest'
        '429':
          description: Daily cap or abuse limit reached.
          content:
            application/json:
              schema:
                $ref: '#/components/schemas/Error'
        '500':
          $ref: '#/components/responses/ServerError'
  /v1/subscriptions:
    post:
      tags: [Subscriptions]
      operationId: createSubscription
      summary: Start a subscription (Phase B)
      description: |
        While the project stays in the SES sandbox, only verified test subscribers receive mail (ADR-0006).
        Stores a pending sign-up in DynamoDB and sends a confirmation message through Amazon SES. The subscriber
receives mail only after confirming (double opt-in).
        Abuse limits are undecided (design-doc OQ-14). Whether the endpoint answers identically for known and unknown
addresses is undecided (OQ-14).
      requestBody:
        required: true
        content:
          application/json:
            schema:
              $ref: '#/components/schemas/SubscriptionRequest'
      responses:
        '202':
          description: Accepted. A confirmation message is sent when applicable.
        '400':
          $ref: '#/components/responses/BadRequest'
        '429':
          description: Too many requests. Limits are not decided (design-doc OQ-14); this response is reserved.
          content:
            application/json:
              schema:
                $ref: '#/components/schemas/Error'
        '500':
          $ref: '#/components/responses/ServerError'
  /v1/subscriptions/confirm:
    get:
      tags: [Subscriptions]
      operationId: confirmSubscription
      summary: Confirm a subscription from the emailed link (Phase B)
      parameters:
        - name: token
          in: query
          required: true
          description: 'Confirmation token: 256-bit, single use, stored as a hash (ADR-0019). Expiry undecided (design-
doc OQ-14).'
          schema:
            type: string
      responses:
        '200':
          description: Subscription confirmed.
        '400':
          $ref: '#/components/responses/BadRequest'
        '404':
          $ref: '#/components/responses/NotFound'
        '500':
          $ref: '#/components/responses/ServerError'
  /v1/subscriptions/unsubscribe:
    get:
      tags: [Subscriptions]
```

```
      operationId: unsubscribePage
      summary: Unsubscribe from the link in an email (Phase B)
      description: Link target. Marks the subscription unsubscribed in DynamoDB without further steps.
      parameters:
        - name: token
          in: query
          required: true
          schema:
            type: string
      responses:
        '200':
          description: Unsubscribed.
        '400':
          $ref: '#/components/responses/BadRequest'
        '404':
          $ref: '#/components/responses/NotFound'
        '500':
          $ref: '#/components/responses/ServerError'
    post:
      tags: [Subscriptions]
      operationId: unsubscribeOneClick
      summary: One-click unsubscribe (Phase B)
      description: Machine request for one-click unsubscribe sent by mail clients. The exact request format is
undecided (design-doc OQ-14).
      parameters:
        - name: token
          in: query
          required: true
          schema:
            type: string
      responses:
        '200':
          description: Unsubscribed.
        '400':
          $ref: '#/components/responses/BadRequest'
        '404':
          $ref: '#/components/responses/NotFound'
        '500':
          $ref: '#/components/responses/ServerError'
components:
  parameters:
    Level:
      name: level
      in: query
      required: false
      description: Reading level. Only standard exists in Phase A; simple ("explain it simply") is added in Phase B
(build plan 2B.2). Value labels proposed (design-doc OQ-22).
      schema:
        type: string
        enum: [standard, simple]
        default: standard
  schemas:
    Brief:
      type: object
      required: [date, level, text, published_at, numbers_verified]
      properties:
        date:
          type: string
          format: date
        level:
          type: string
          enum: [standard, simple]
        text:
          type: string
          description: The plain-language brief. Every number in it is in the tables.
        published_at:
          type: string
          format: date-time
          description: UTC.
        numbers_verified:
          type: boolean
          description: Always true on a published brief.
        closest_approaches:
          type: array
          description: The facts the text states. Item fields pending (design-doc OQ-1).
          items:
```

```
          type: object
      fireballs:
        type: array
        description: The day's fireballs for the map. Item fields pending (OQ-1).
        items:
          type: object
      changed_predictions:
        type: array
        description: Predictions that changed since yesterday. Item fields pending (OQ-1).
        items:
          type: object
      share_card_url:
        type: string
        format: uri
        description: Phase B (build plan 2B.2). Format undecided (design-doc OQ-11).
  BriefIndex:
    type: object
    required: [items]
    properties:
      items:
        type: array
        items:
          type: object
          required: [date]
          properties:
            date:
              type: string
              format: date
            levels:
              type: array
              items:
                type: string
  CloseApproachList:
    type: object
    required: [items]
    properties:
      items:
        type: array
        description: Item fields pending (design-doc OQ-1).
        items:
          type: object
  FireballList:
    type: object
    required: [items]
    properties:
      items:
        type: array
        description: Item fields pending (design-doc OQ-1).
        items:
          type: object
  PredictionHistory:
    type: object
    required: [designation, versions]
    properties:
      designation:
        type: string
      versions:
        type: array
        description: SCD Type 2 rows, oldest first.
        items:
          type: object
          required: [valid_from_pull_time]
          properties:
            valid_from_pull_time:
              type: string
              format: date-time
            valid_to_pull_time:
              type: string
              format: date-time
              nullable: true
            predicted_values:
              type: object
              description: Fields pending (design-doc OQ-1).
      flags:
        type: array
        description: Shift flags with explanations (build plan 2B.3).
```

```
        items:
          type: object
          properties:
            flagged_at_pull_time:
              type: string
              format: date-time
            explanation:
              type: string
  HistoryQuestionRequest:
    type: object
    required: [question]
    properties:
      question:
        type: string
        description: Free-text question. Maximum length undecided (design-doc OQ-17).
  HistoryAnswer:
    type: object
    required: [answer, source]
    properties:
      answer:
        type: string
      source:
        type: string
        enum: [metric, sql_fallback]
        description: Whether a defined metric or the logged SQL fallback produced the result.
      result:
        type: array
        description: The query result the answer is based on.
        items:
          type: object
  SubscriptionRequest:
    type: object
    required: [email]
    properties:
      email:
        type: string
        format: email
        description: Stored only in DynamoDB (ADR-0007).
      level:
        type: string
        enum: [standard, simple]
        description: Whether subscribers choose a level is not stated in the plan (design-doc OQ-13).
  Error:
    type: object
    description: Error body shape is proposed (design-doc OQ-22).
    required: [error]
    properties:
      error:
        type: object
        required: [code, message]
        properties:
          code:
            type: string
          message:
            type: string
responses:
  BadRequest:
    description: The request is malformed or a parameter is invalid.
    content:
      application/json:
        schema:
          $ref: '#/components/schemas/Error'
  NotFound:
    description: Nothing exists for that identifier, date or token.
    content:
      application/json:
        schema:
          $ref: '#/components/schemas/Error'
  ServerError:
    description: Unexpected error.
    content:
      application/json:
        schema:
          $ref: '#/components/schemas/Error'
```

Part 10: Changelog

Asteroid Watch/changelog.md

Changelog: Project 2, Asteroid Watch

Dates are the day of the change. Counts are checked against the open-questions table in design-doc.md section 16.

Oct 9, 2026: defaults confirmed

- John confirmed ADR-0015 (expected signature version per API), in addition to ADRs 0016 to 0019 confirmed earlier. Their status is now Accepted.

Oct 9, 2026: revision 4 (defaults confirmed)

- John confirmed four defaults: ADR-0016 (SHA-256 content hash), ADR-0017 (HMAC-SHA-256 keyed hash, read by scan), ADR-0018 (fail closed on upstream failures and signature change) and ADR-0019 (single-use hashed confirmation tokens). Their status is now Accepted.
- OQ-4 and OQ-5 are answered. OQ-14 and OQ-15 stay partly answered (expiry, rate limits, key rotation and retention are still open).

Oct 9, 2026: revision 3 (while-building documents added)

- Added README, operations, model card, and security and privacy for Project 2. These are the while-building documents in the Documentation Plan, written before coding so the rules are fixed in advance.
- The model card is a draft: the model is not named (OQ-8) and no evals have run.
- The security document lists the deletion periods that John must set (OQ-15, OQ-14, OQ-19). These are decisions, not defaults.
- Source checks added in the security document: SES sandbox limits (verified), JPL terms (unverified, OQ-24).
- Reference generated from code is not written yet: it needs code. Planned after the first pipeline exists.
- Open-question counts are unchanged in this revision: 26 in total, 1 answered by default, 7 partly answered, 18 open.

Oct 9, 2026: revision 2 (answered decisions applied; checks and fixes)

Answered decisions written into the documents

- Late data shows the old data with a late notice (ADR-0013, FR-43). Already in revision 1; cross-references checked.
- SES stays in the sandbox (ADR-0006). The SES limits (verified recipients only, 200 messages a day, 1 a second) were checked against AWS's documentation.
- History questions are public with a cap, a token limit and abuse limits (ADR-0002, ADR-0009). Numbers still open (OQ-17).
- The platform defines the usage-event format and collection point (ADR-0010). Delete-window length and session definition still open (OQ-19).
- The pulls log plus distinct-versions design is approved (ADR-0003). The first-run test case is named (ADR-0014).

New ADRs

- ADR-0014: first-run test case 2020 QG, found in the live API. Accepted (John's answer 39).
- ADR-0015: expected signature version per API (1.5 close approaches, 1.2 fireballs). Default.
- ADR-0016: content hash is SHA-256 of raw bytes. Default; answers OQ-4.
- ADR-0017: subscriber copy uses HMAC-SHA-256 and a scan, not an export. Default.
- ADR-0018: failed or malformed upstream responses fail with an alert; a signature change keeps the raw file and stops publishing. Default.

- ADR-0019: confirmation and unsubscribe tokens are 256-bit, stored as hashes; confirmation tokens are single use. Default.

Corrections found in checking

- The fireball API reports signature 1.2, not 1.5 (checked Oct 9, 2026). The one-value rule would have alerted on every fireball run. Fixed in FR-4, QR-13, ADR-0004 and both contracts.
- The subscriber level question cited OQ-12 (the RSS question). It now cites OQ-13 in the subscriber contract and the OpenAPI file.
- An unquoted YAML value in openapi.yaml had a colon followed by a space. Quoted; the file parses.
- The DynamoDB export would have put plain addresses in S3. The copy now uses a scan (ADR-0017).
- The open-questions table keeps every row and adds a Status column with the answer and ADR number. Section 16's rule about removing answered rows was changed to match.
- A glossary entry for HMAC-SHA-256, SHA-256 and fail closed was added.

Verified against primary sources (Oct 9, 2026)

- JPL close-approach API: 2020 QG is present for August 2020; signature 1.5 at `signature.version`.
- JPL fireball API: signature 1.2 at `signature.version`; 237 records in the sample.
- JPL close-approach API page: no terms, license or credit line stated. JPL's and NASA's policy pages returned 404; JPL terms remain unverified (OQ-24).
- AWS SES sandbox documentation and AWS DynamoDB export documentation.

Counts (section 16)

- Open questions: 26 in total. 1 answered by default (OQ-4). 7 partly answered (OQ-1, OQ-5, OQ-13, OQ-14, OQ-15, OQ-17, OQ-19). 18 open, including OQ-2, whose test case is now named.
- Requirements: 45 functional items (FR-1 to FR-43, with FR-32 split into three), 21 quality scenarios (QR-1 to QR-21).
- ADRs: 19 (ADR-0001 to ADR-0019). ADR-0012 is still Proposed.

Still for John

- Confirm the defaults in ADR-0015, 0016, 0017, 0018 and 0019, and the reading of "1.5" as the close-approach value.
- Decide OQ-9 (late threshold), OQ-10 (static files or Lambda), OQ-26 (brief date) and the other open items in section 16.

Revision 1 (Oct 9, 2026)

- First full set: requirements, design doc, OpenAPI, 13 ADRs, 12 contracts.